

Past, Present, and Future Climate



What we have accomplished

Chapter 1: Astronomical factors

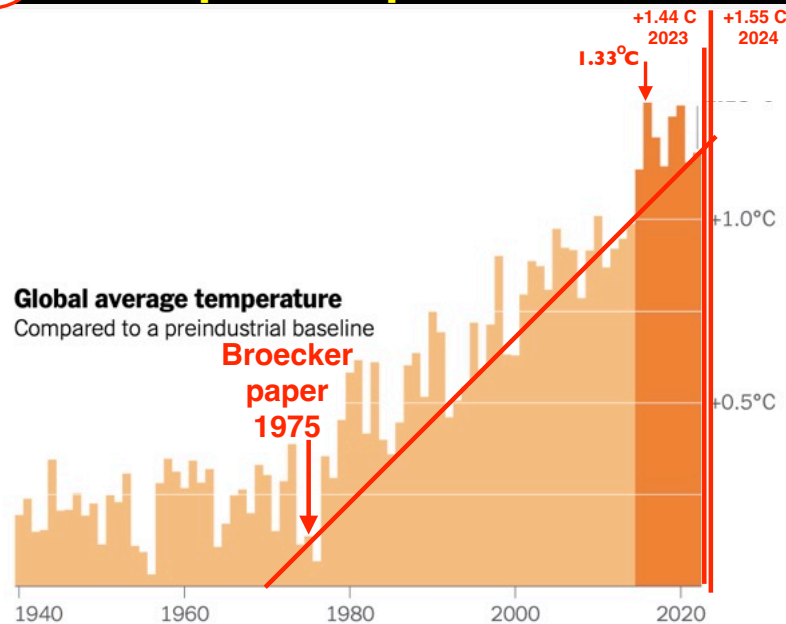
Chapter 2: Earth's surface and atmosphere

Chapter 3: The source of CO₂

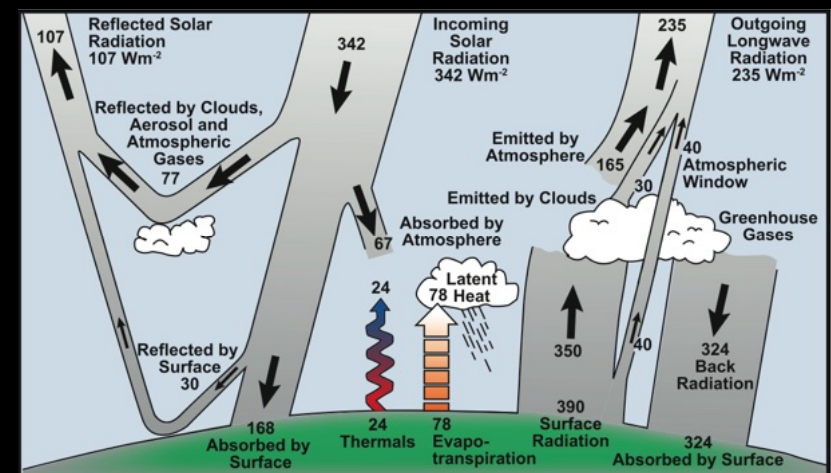
Chapter 4: The temperature record

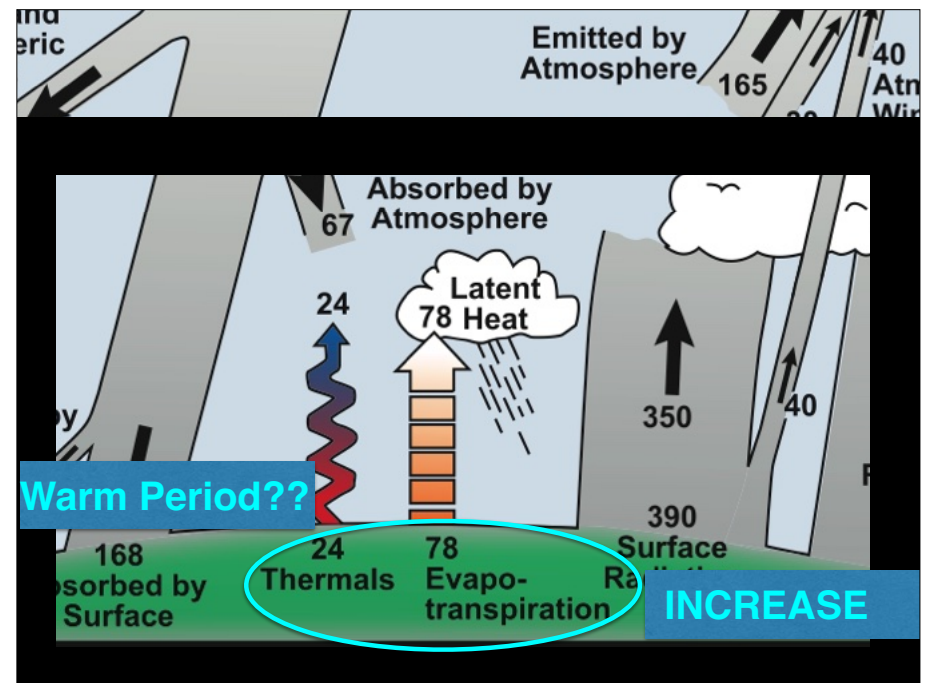
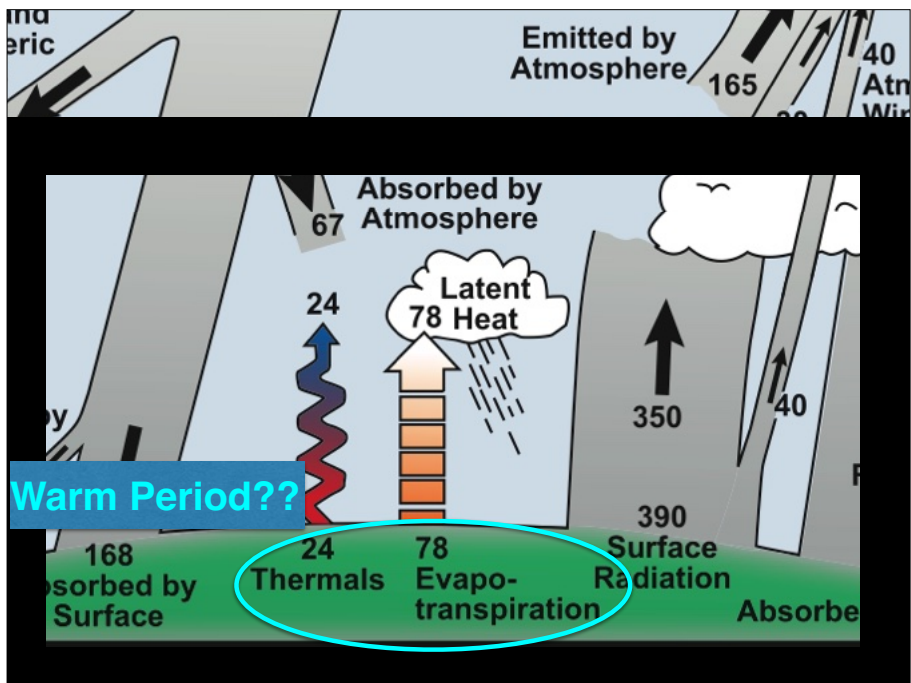
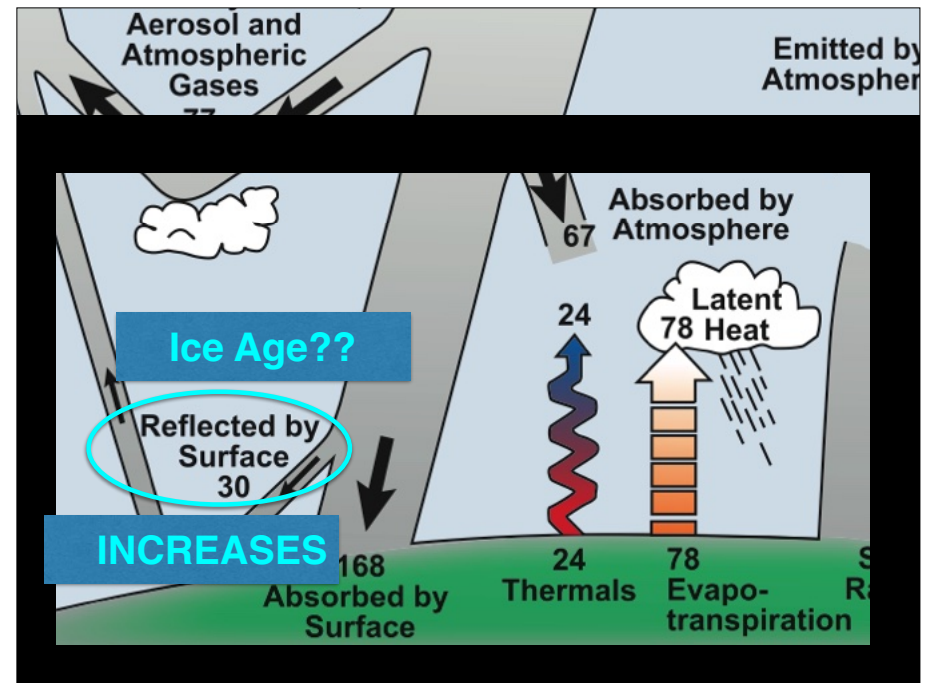
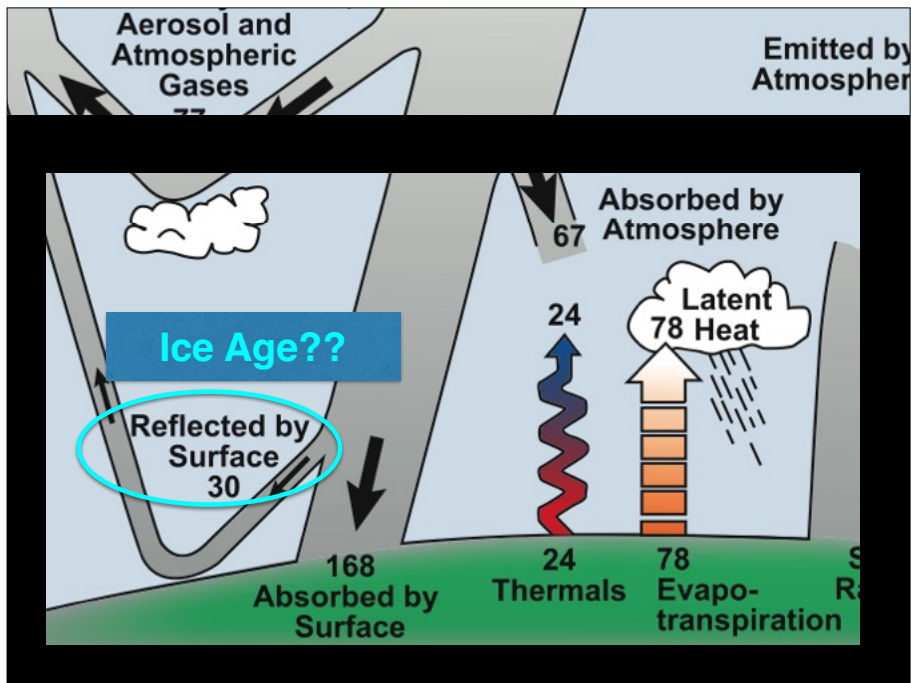
Fact

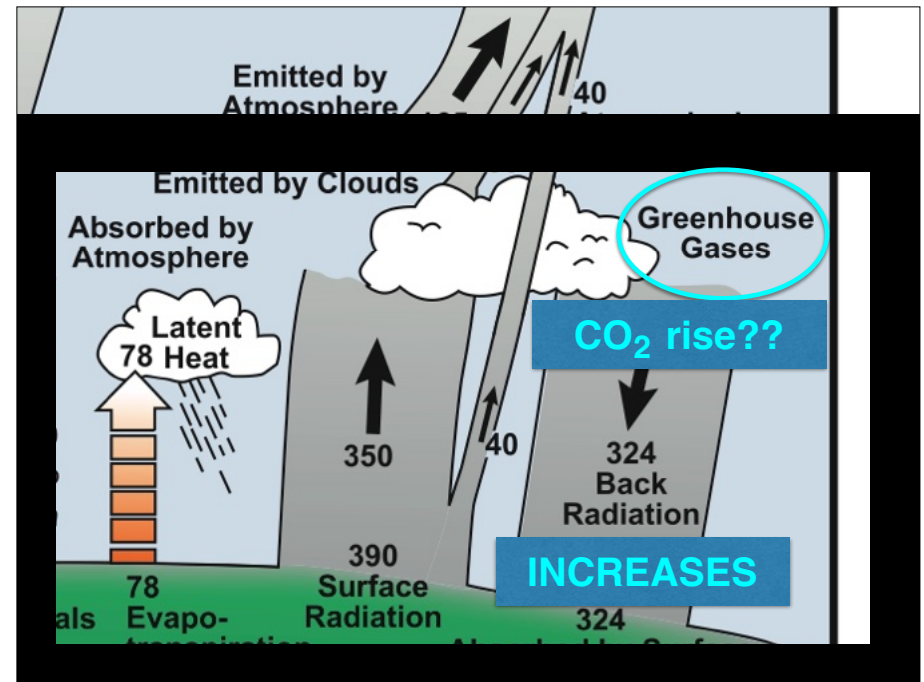
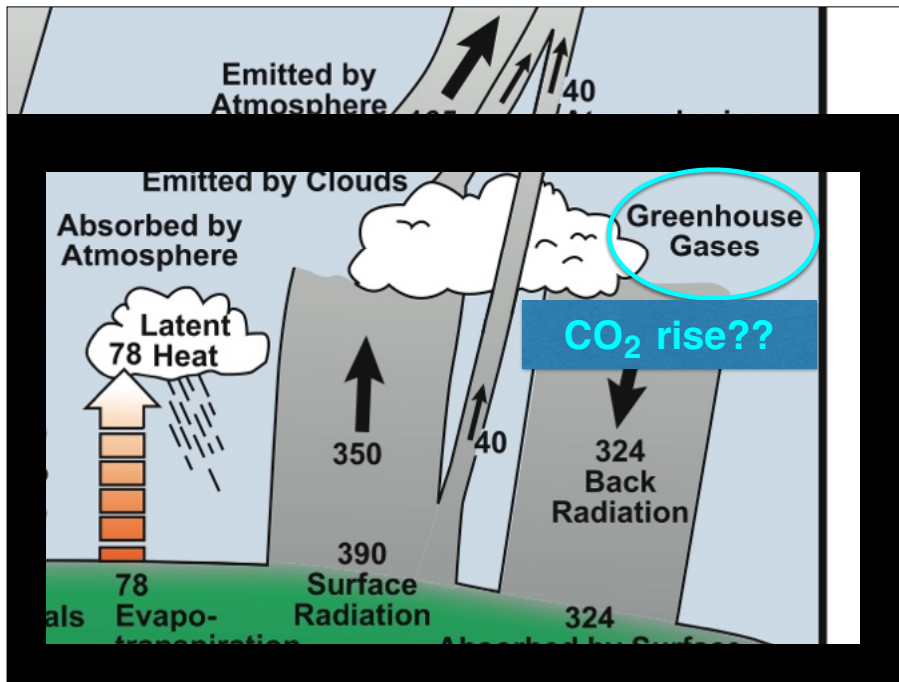
T compared to pre-industrial T



Energy IN = Energy OUT







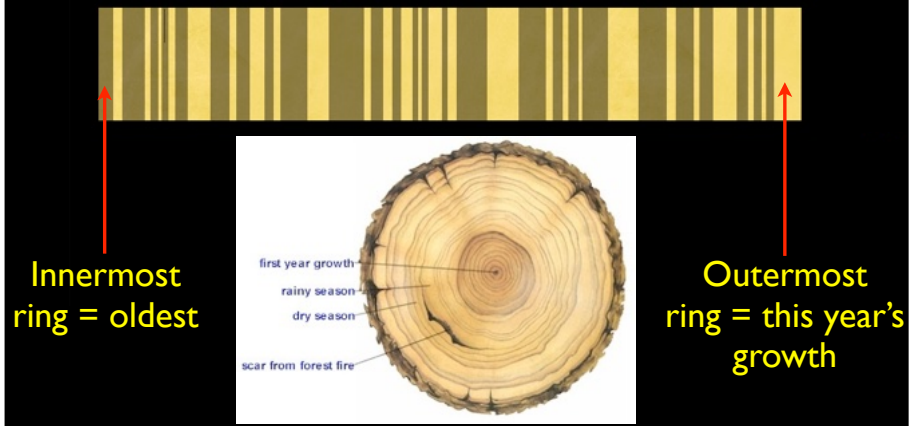
Chapter 5: Temperature over long time scales

Climate Records

Record	Timescale
Instrumental measurements	<150 yrs
Written historical records	~2500 yrs
Tree rings	13,910 yrs
Ice Cores	800,000 yrs
Ocean floor sediments	50 million yrs
Fossils and Rocks	4.4 billion yrs

Dendrochronology

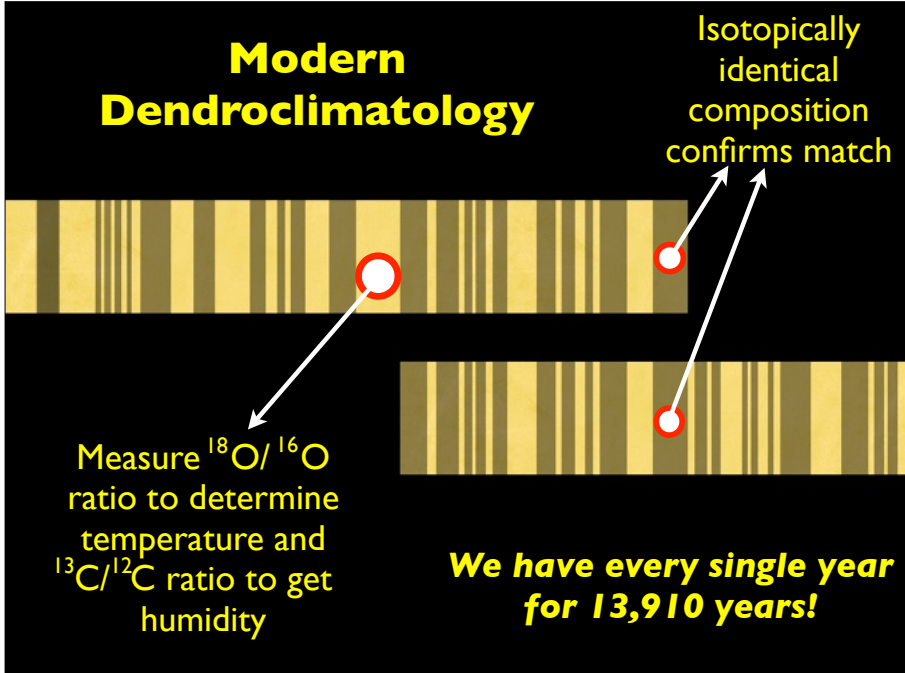
One can “read the history of past seasons in tree rings” --Leonardo da Vinci



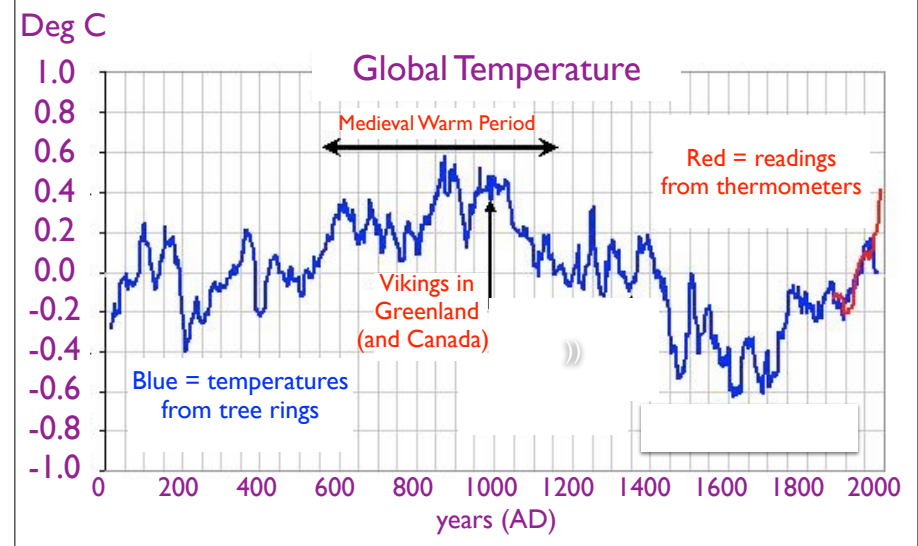
Bristlecone Pines: The oldest living things on Earth

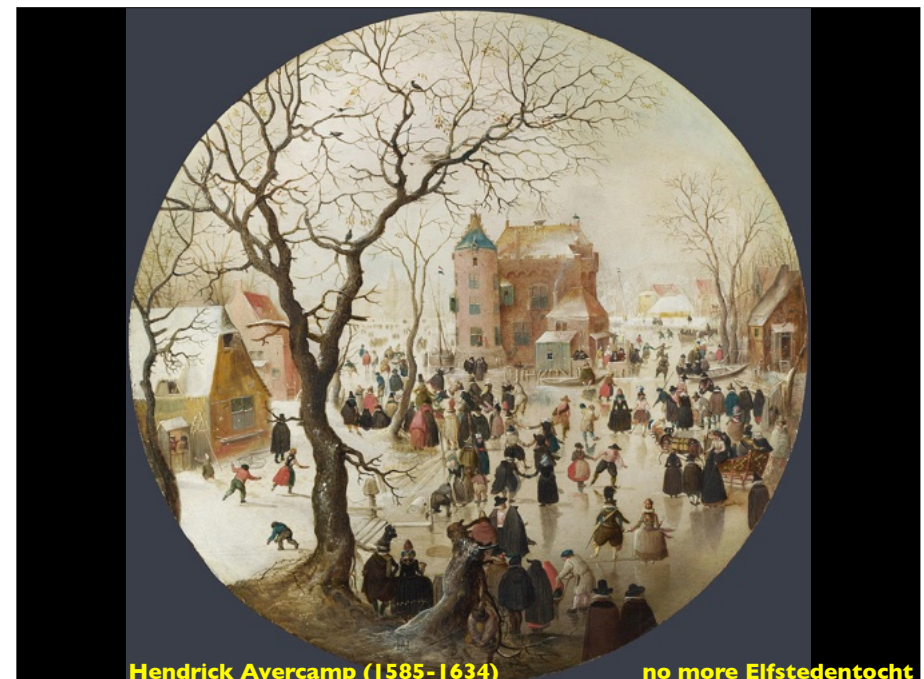
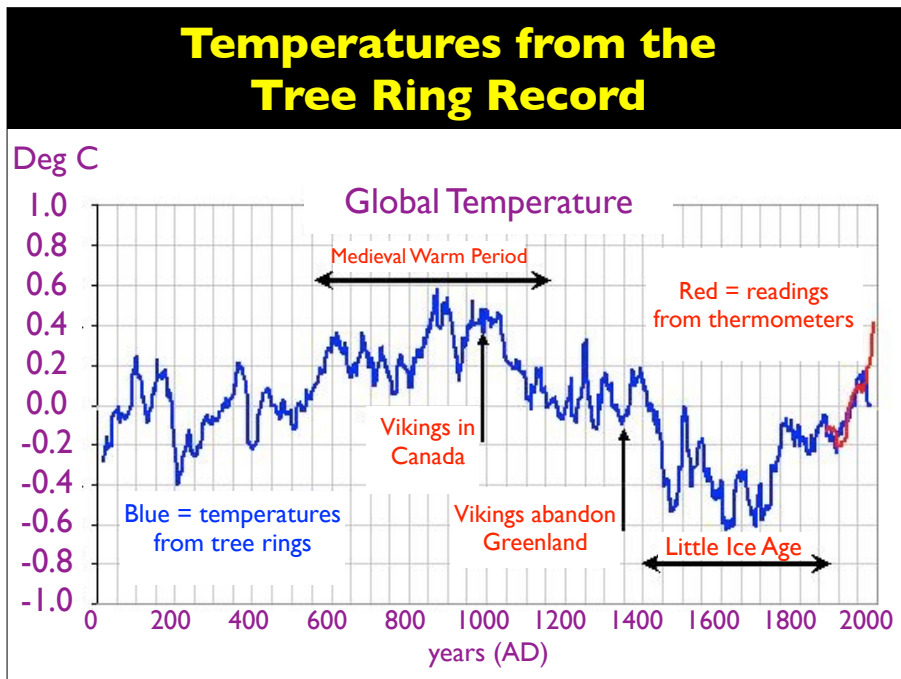
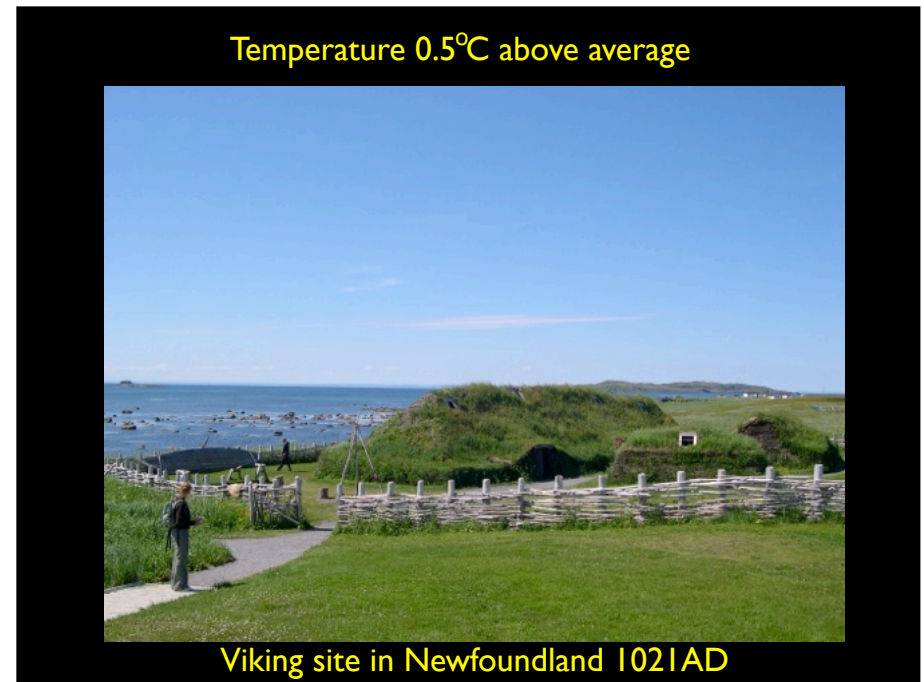


Modern Dendroclimatology

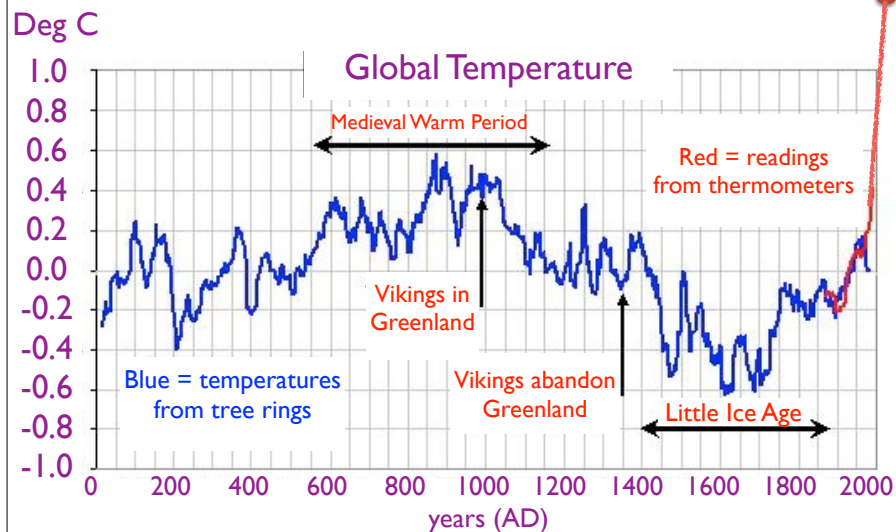


Temperatures from the Tree Ring Record

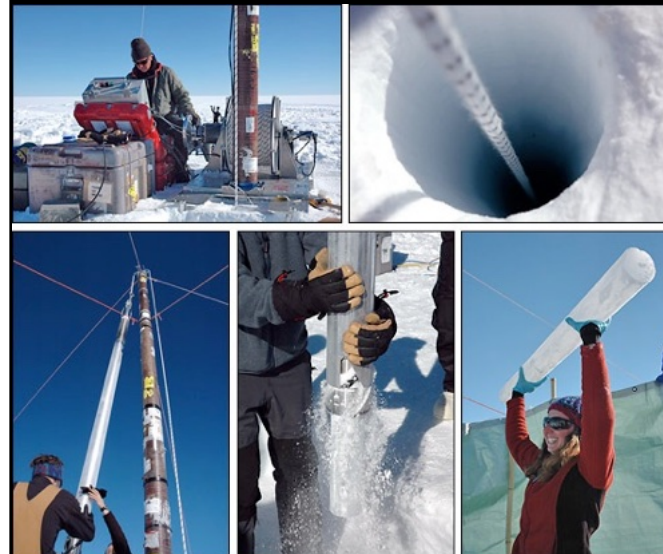




"the exponential rise in the atmospheric carbon dioxide content ... will have driven the mean planetary temperature beyond the limits experienced during the last 1000 years." (Broecker 1975)



Ice Cores from Greenland and Antarctica

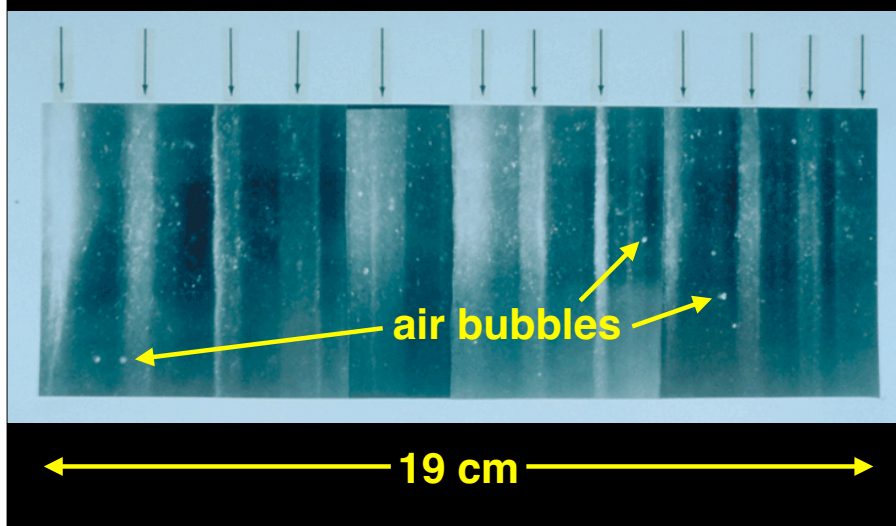


Isotopes give temperature

Air bubbles give atmospheric samples

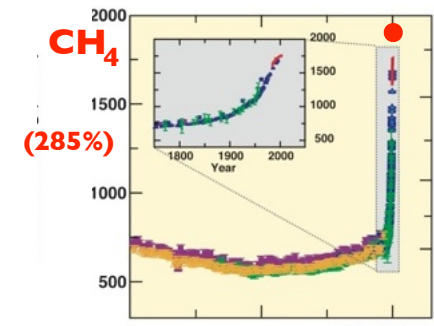
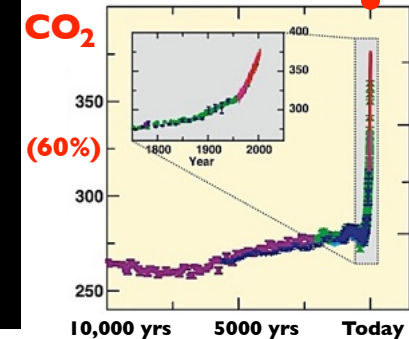
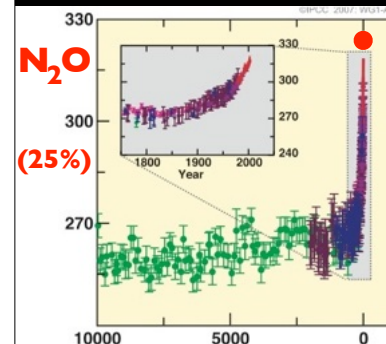
Salt and dust give wind speed

Ice core from 1.855 km deep in Greenland still shows 12 annual layers

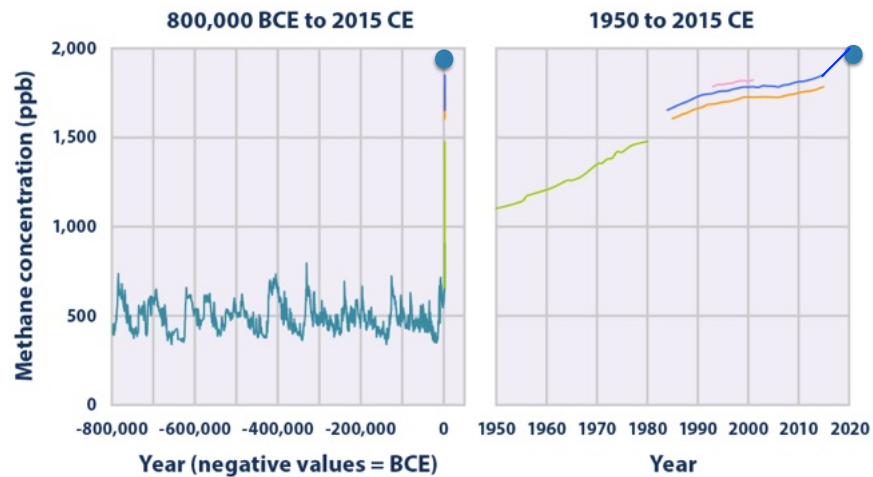


Fact

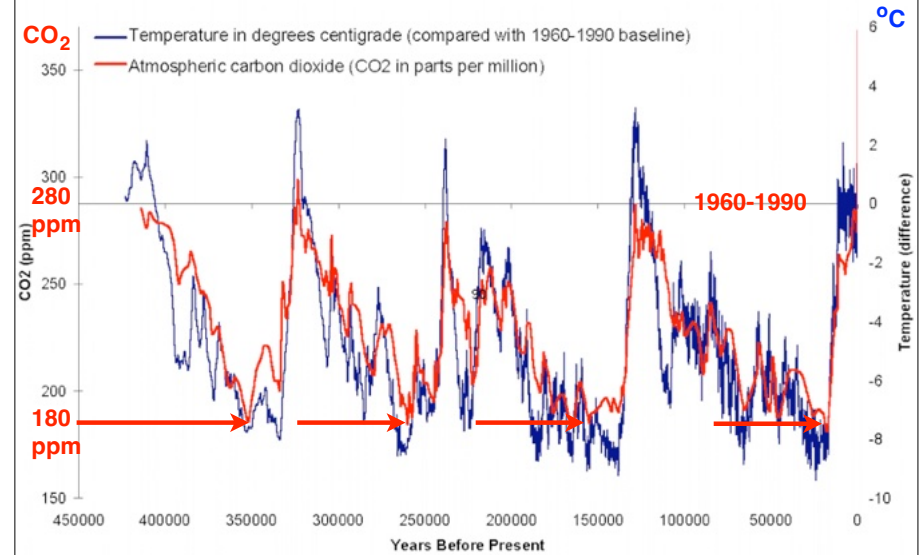
Carbon dioxide, Methane, and nitrogen oxides have all risen rapidly since 1800



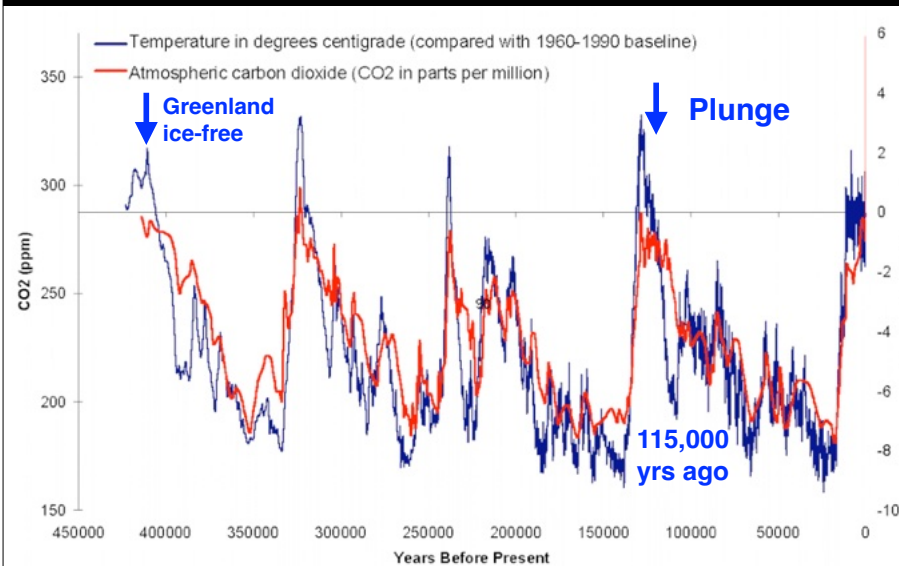
Methane over the last ~1 million years



Fact 400,000 years of CO₂ and Climate

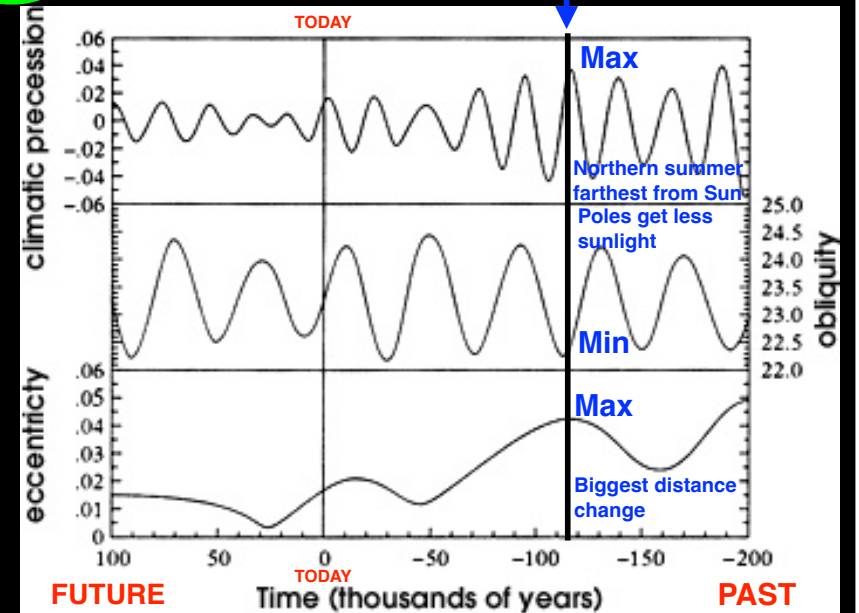


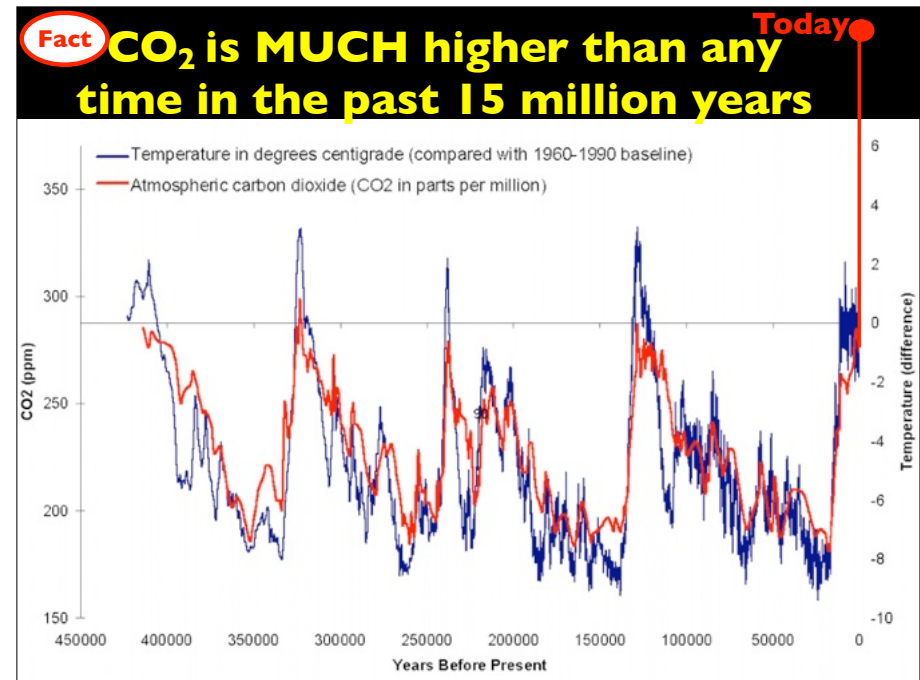
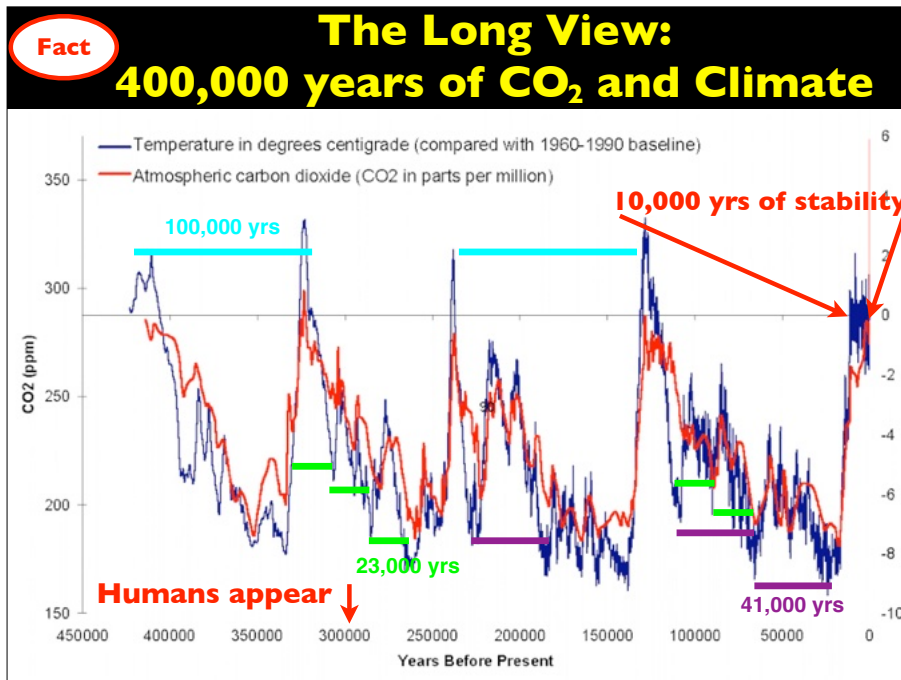
Fact 400,000 years of CO₂ and Climate



Physics

Plunge into last Ice Age





Chapter 5: Paleoclimatology

- we have excellent proxies for past climate
- we see the signatures of orbital changes in past climate
- Greenhouse gases have all increased dramatically since the dawn of the Industrial Revolution
- CO₂ and temperature are highly correlated in the past

Chapter 6: Energy balance and feedback

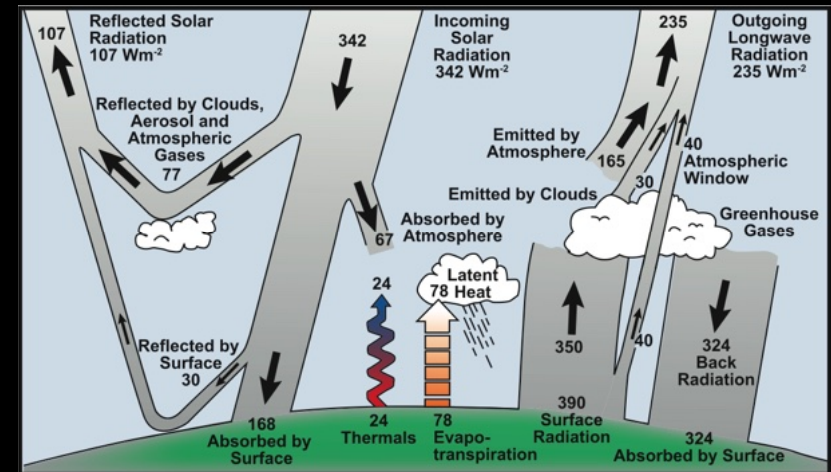
The Past is the key to the Future

With all these **Facts** and all this **Physics**
we can build a MODEL
and try to predict
future climate

It's just

$$\text{Energy IN} = \text{Energy OUT}$$

$$\text{Energy IN} = \text{Energy OUT}$$



Feedback

But it is **VERY** hard because of **FEEDBACK**

Temperature $\uparrow \Rightarrow$ Arctic ice cover $\downarrow \Rightarrow$ light reflected $\downarrow \Rightarrow$
light absorbed $\uparrow \Rightarrow$ ocean gets warmer $\Rightarrow$ Temperature

Positive Feedback: The effect amplifies the cause

Feedback

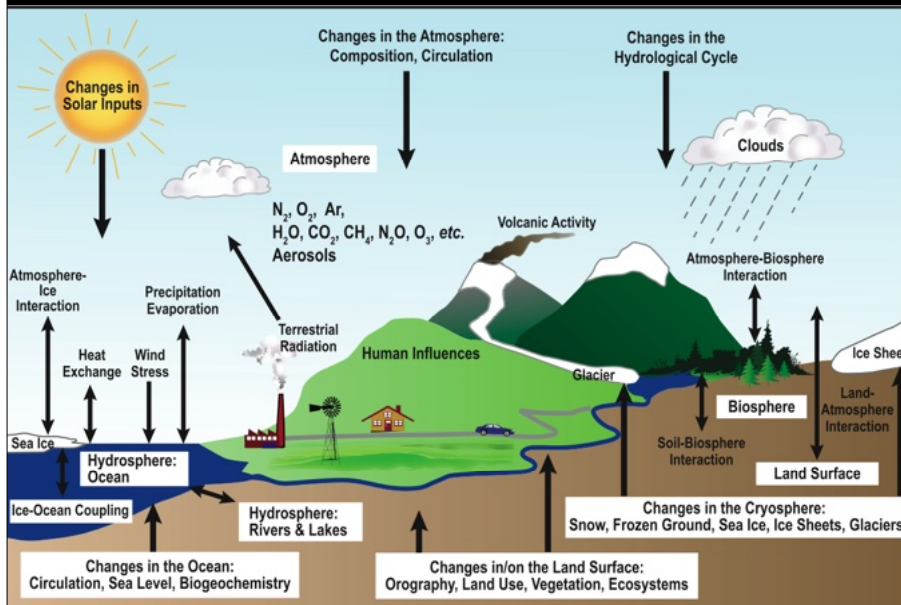
But it is **VERY** hard because of **FEEDBACK**

Temperature $\uparrow \Rightarrow$ Arctic ice cover $\downarrow \Rightarrow$ light reflected $\downarrow \Rightarrow$
light absorbed $\uparrow \Rightarrow$ ocean warmer $\Rightarrow$ Temperature
 $\Rightarrow$ evaporation $\uparrow \Rightarrow$ clouds $\Rightarrow$ reflected light $\uparrow$

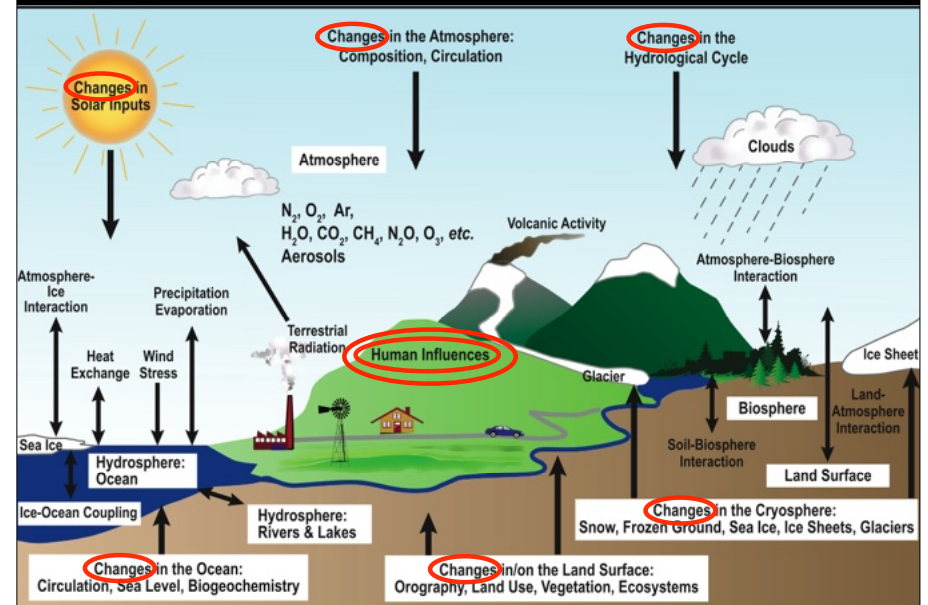
Negative Feedback: The effect suppresses the cause

Recent study: 27 positive, 7 negative, 7 uncertain

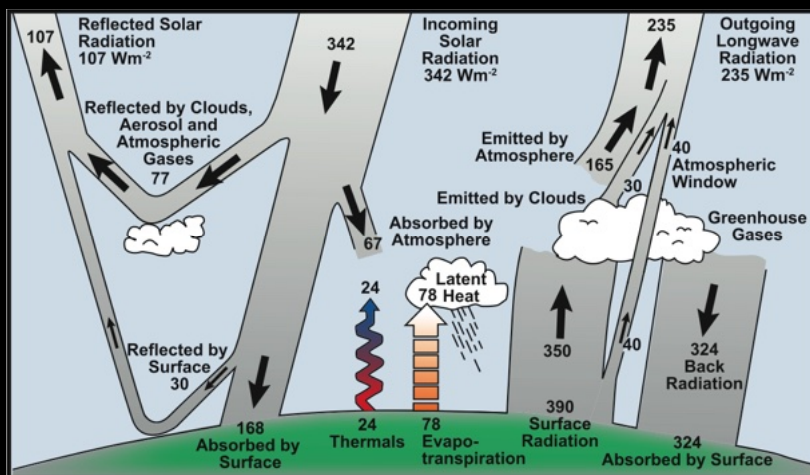
Positive and Negative Feedbacks Abound



and all parts are constantly changing



Change one of the values in this diagram and think about what feedback loops this might trigger



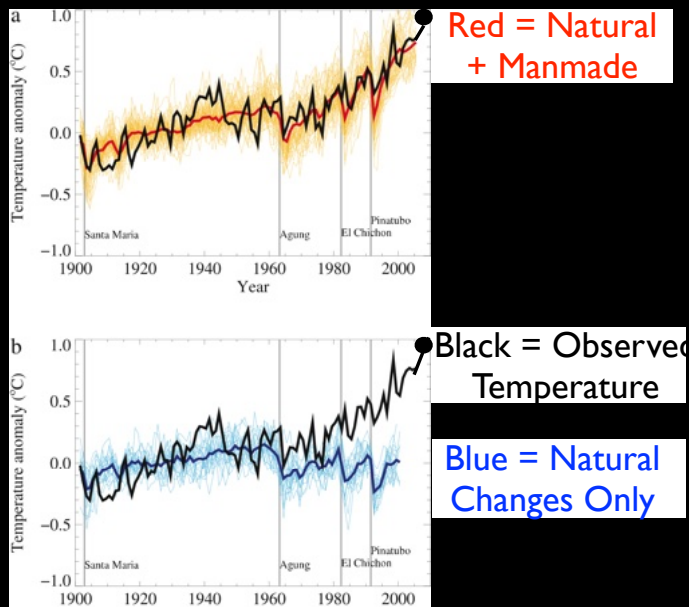
Chapter 6: Energy balance and feedback

- positive feedback: the effect amplifies the cause
- negative feedback: the effect suppresses the cause
- the climate system has dozens of feedback loops
- coupled feedback loops make modeling challenging

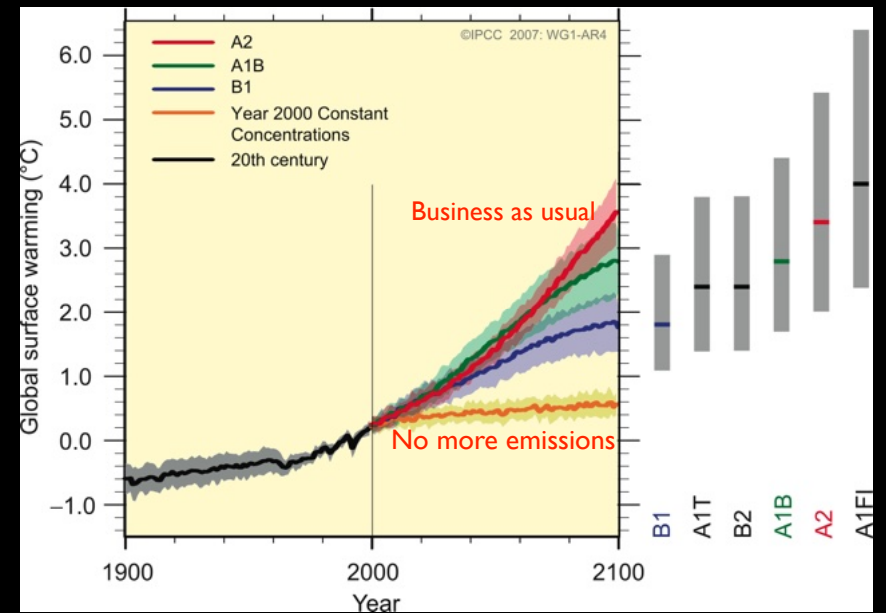
Chapter 7: Modeling the climate

Can computer models hope to cope?

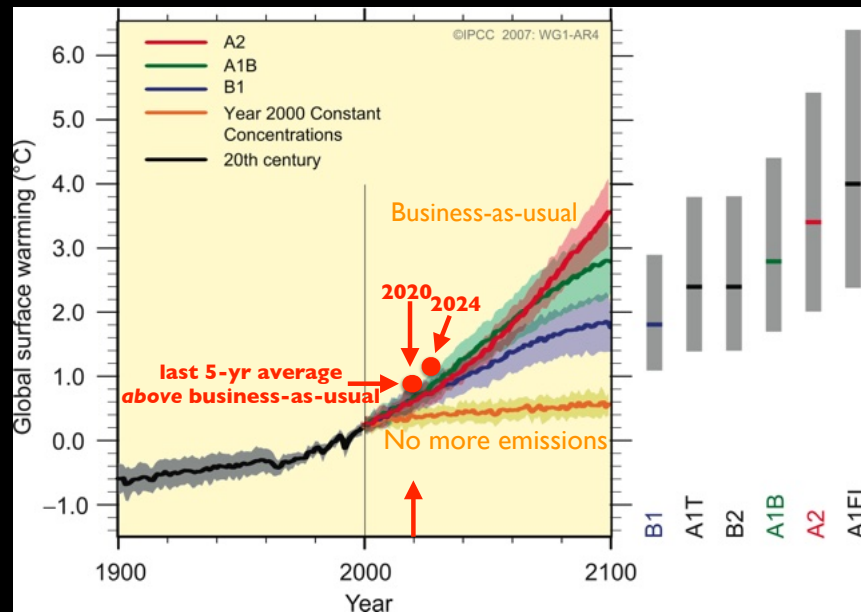
The models work reasonably well



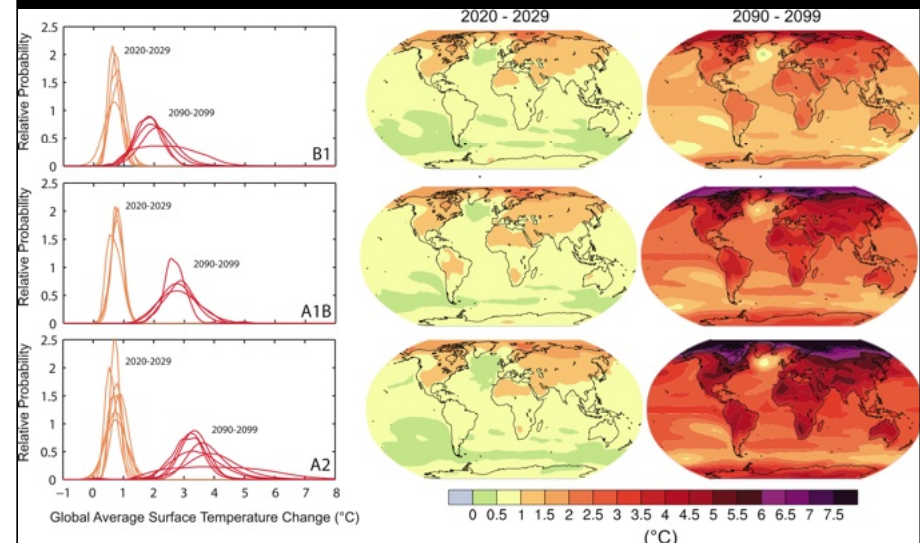
And here is what they predict



And here is what they predict



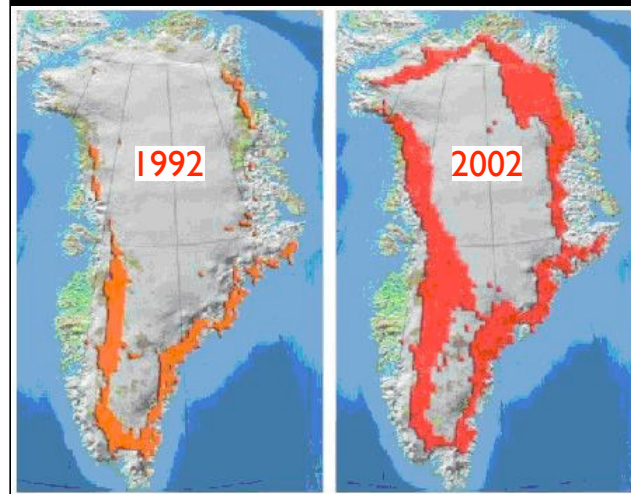
And different models agree reasonably well



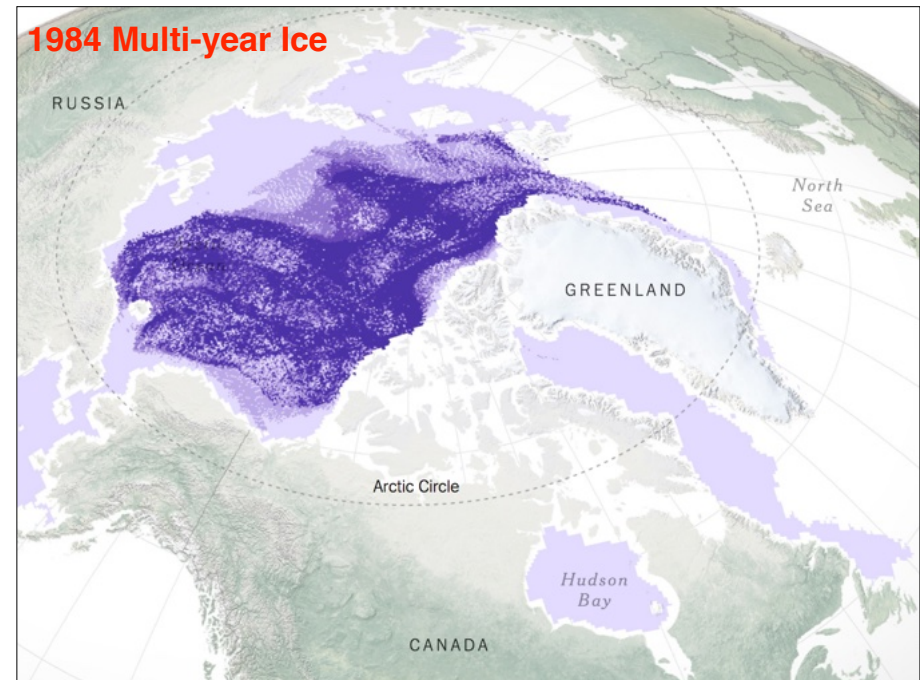
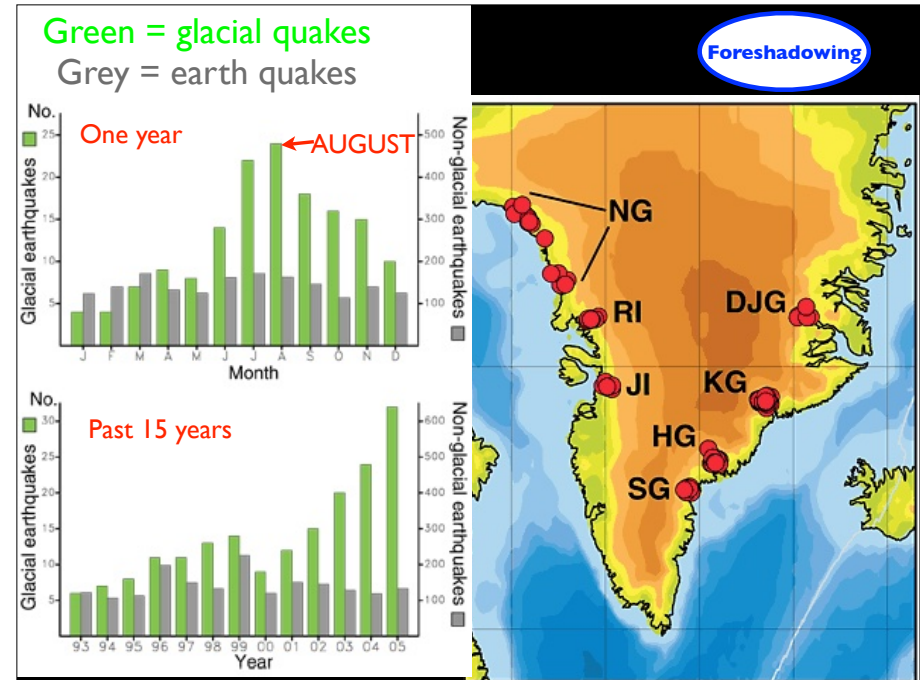
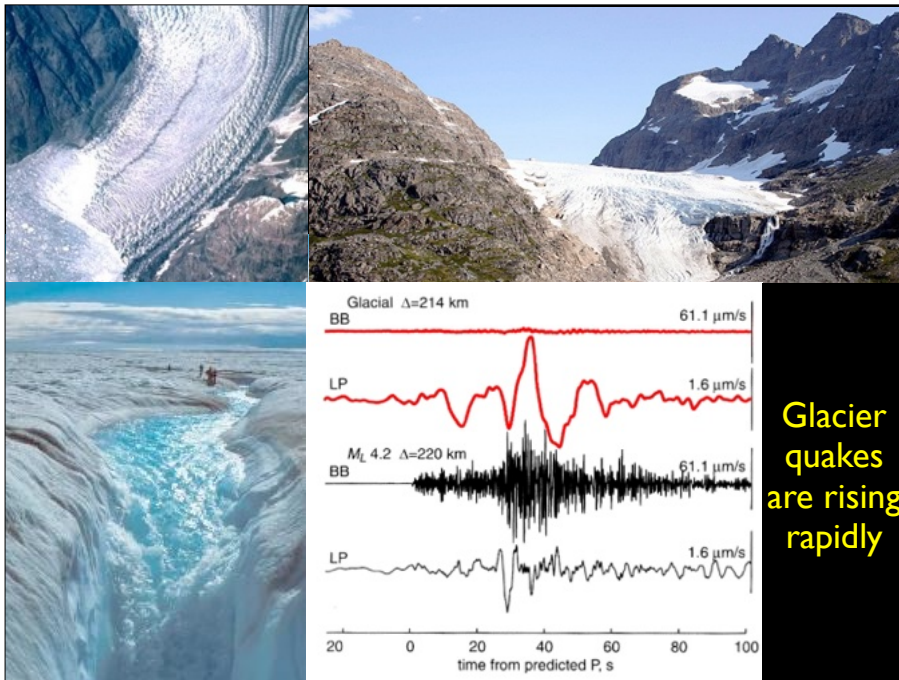
Are we confident of these predictions?

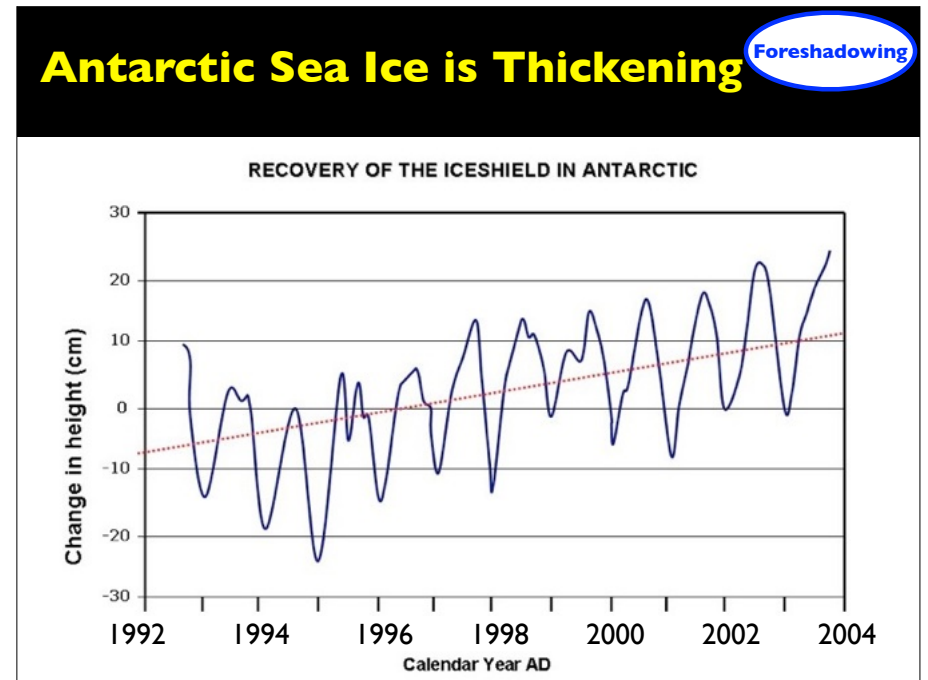
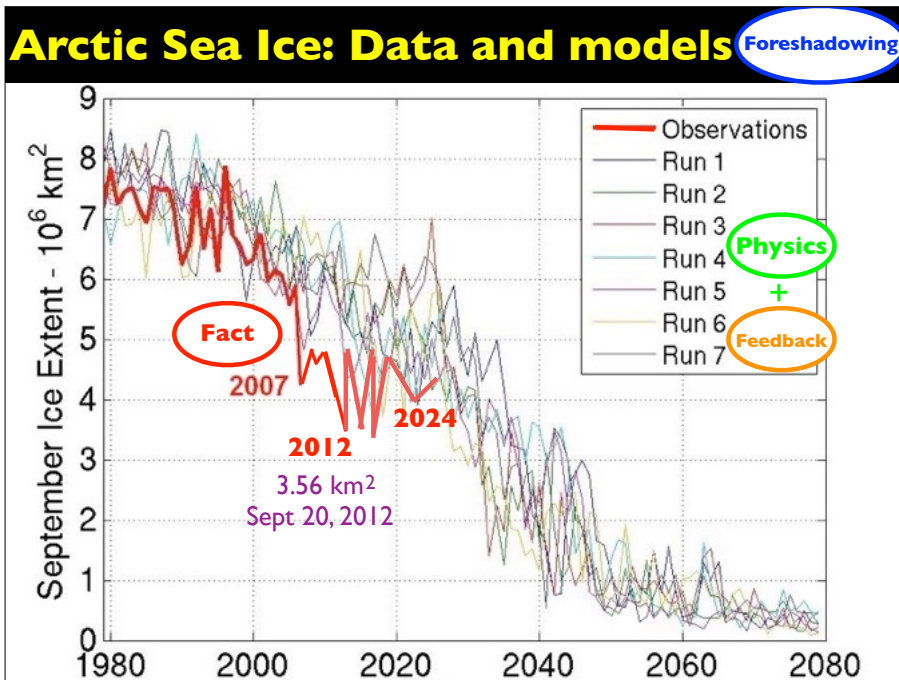
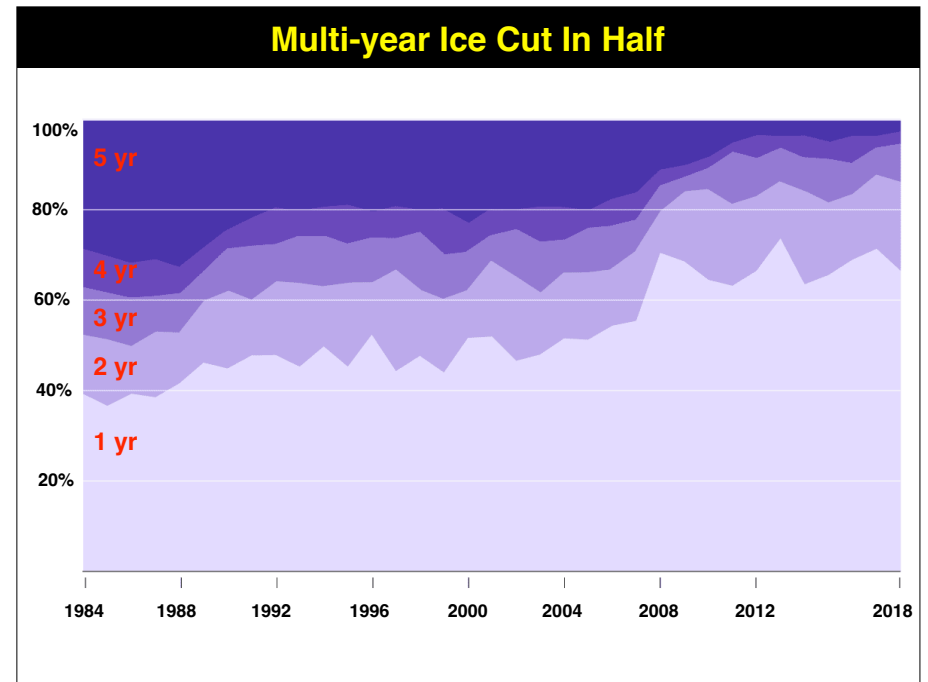
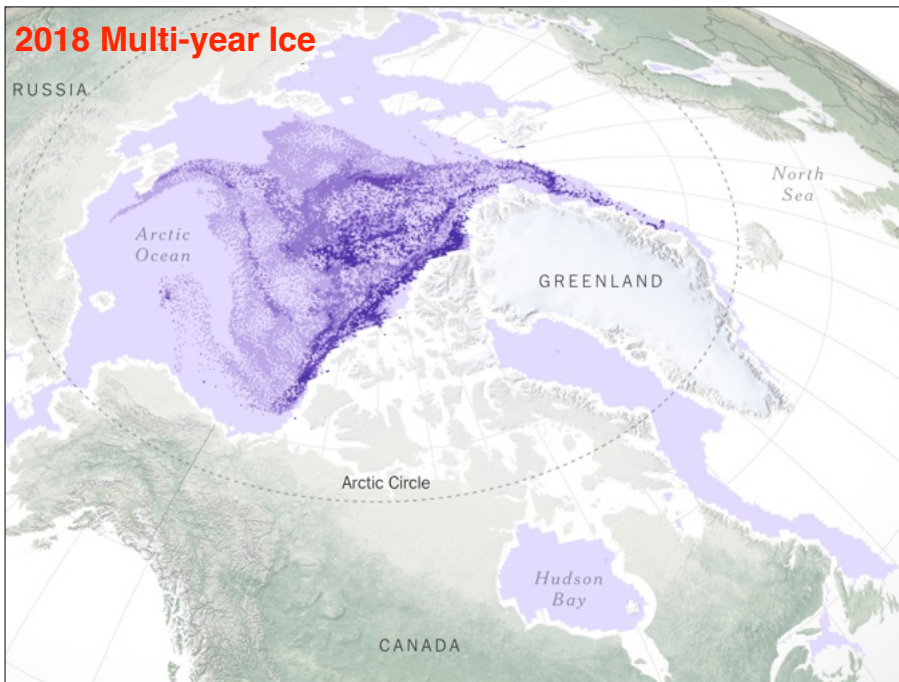
Greenland's Ice Cap

Foreshadowing



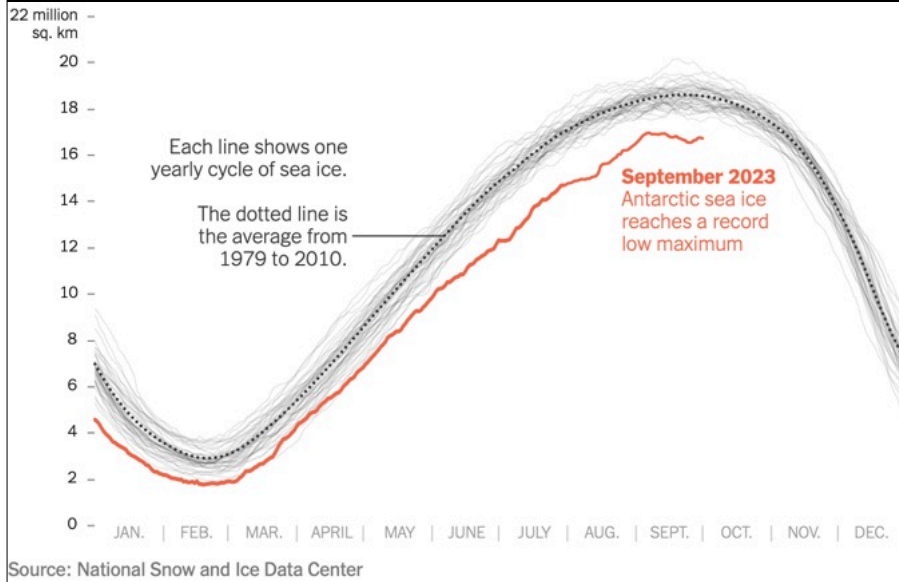
- 700,000 mi²
- 2 miles thick
- glacial flow rate doubled in 5 yrs
- air temperature up 4°F in 20 yrs
- sea level rise if melted = 23 ft





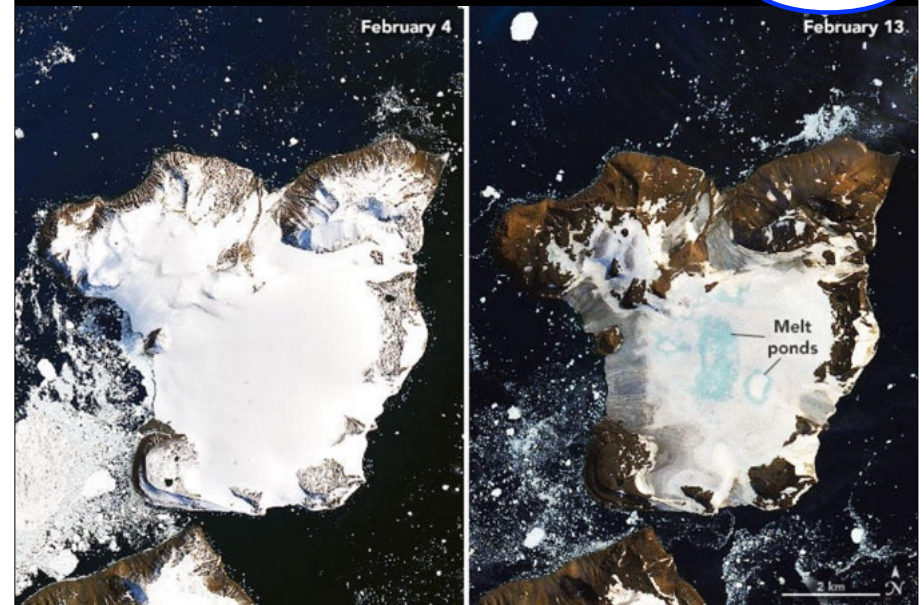
Not any more...

Foreshadowing



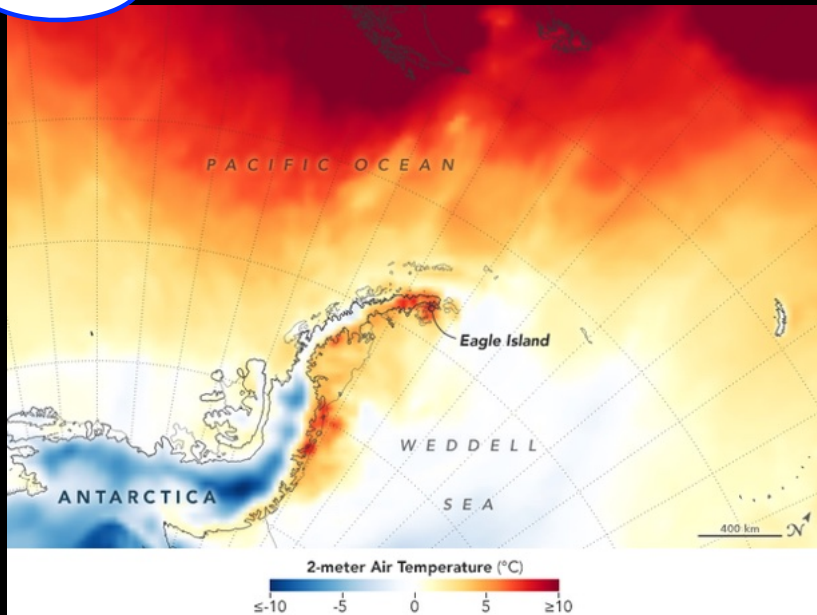
2020

Foreshadowing

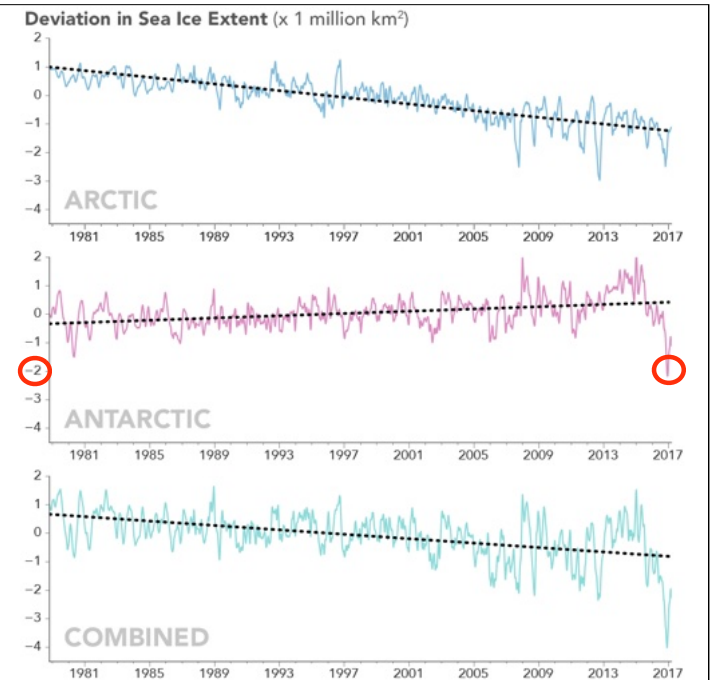


Foreshadowing

2020 Feb 6: T=64.9 F

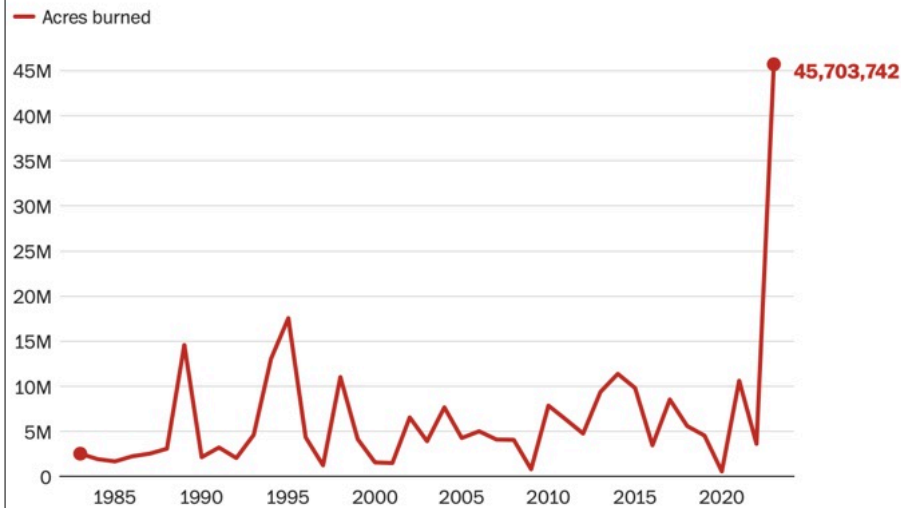


Until recently



Canadian Forest Fires 9 times the average of the past 40 years

An unprecedented year for wildfire in Canada



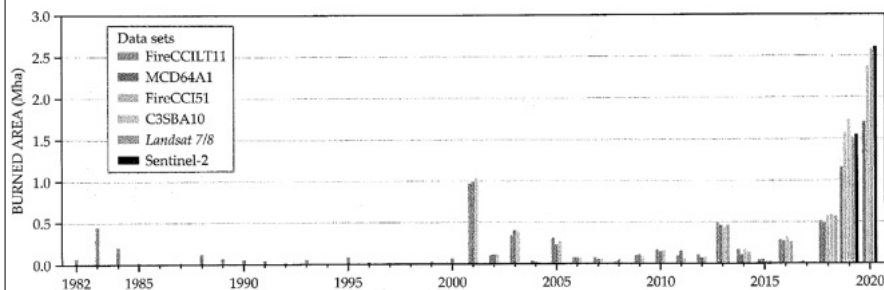
Foreshadowing

Siberian Forest Fires

1982-2018: 0.3 million acres/yr
2019-2020: 11.5 million acres/yr (0.4 Gtons CO₂)



Siberian Forest Fires Million hectares burned over past 40 years

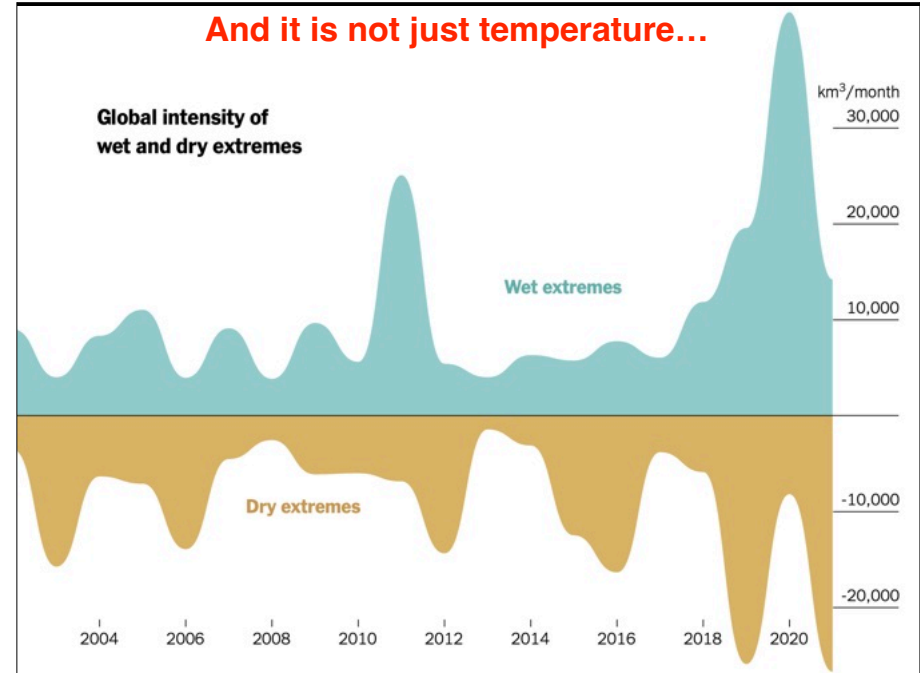


Higher T $\Rightarrow$ earlier snow melt $\Rightarrow$ lower reflectivity
 $\Rightarrow$ higher T $\Rightarrow$ longer growing season $\Rightarrow$ insufficient water $\Rightarrow$ higher fire risk

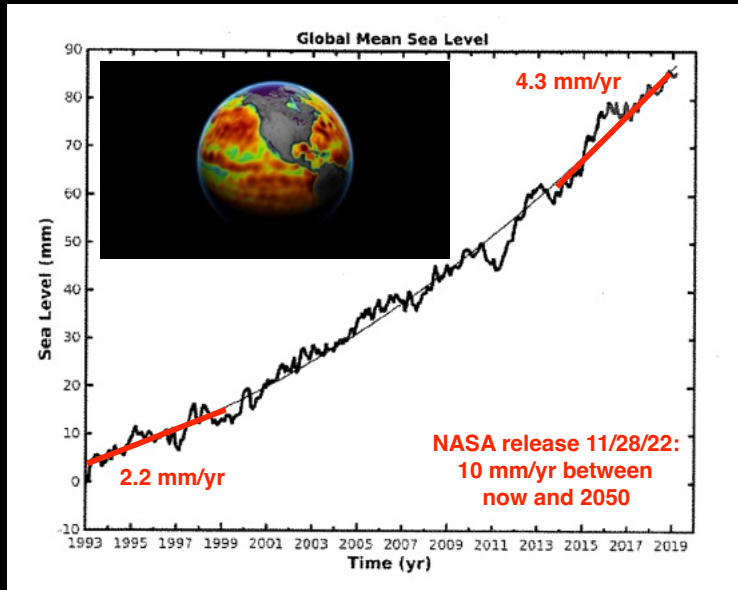
Boreal fires produced 23% of land-use emissions in 2021
 Re-forestation absorbs only 26% of CO₂ from deforestation

And it is not just temperature...

Global intensity of wet and dry extremes

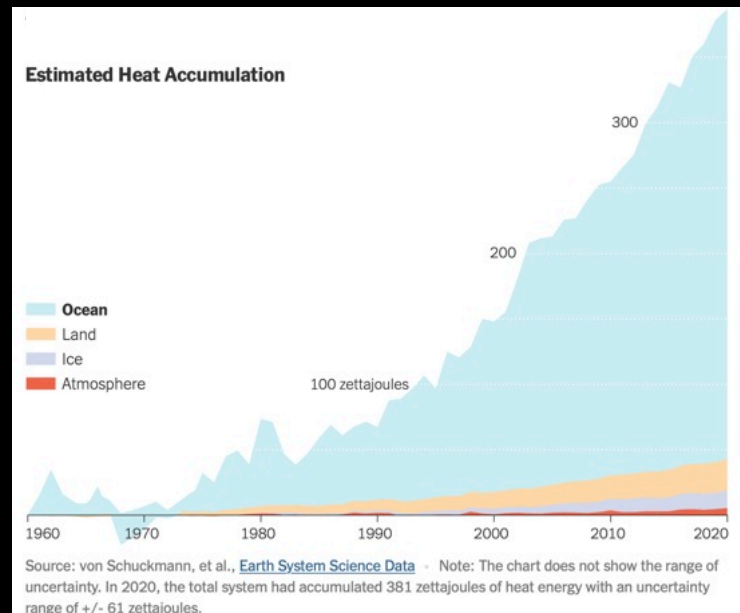


Sea Level Rise 1993-2019

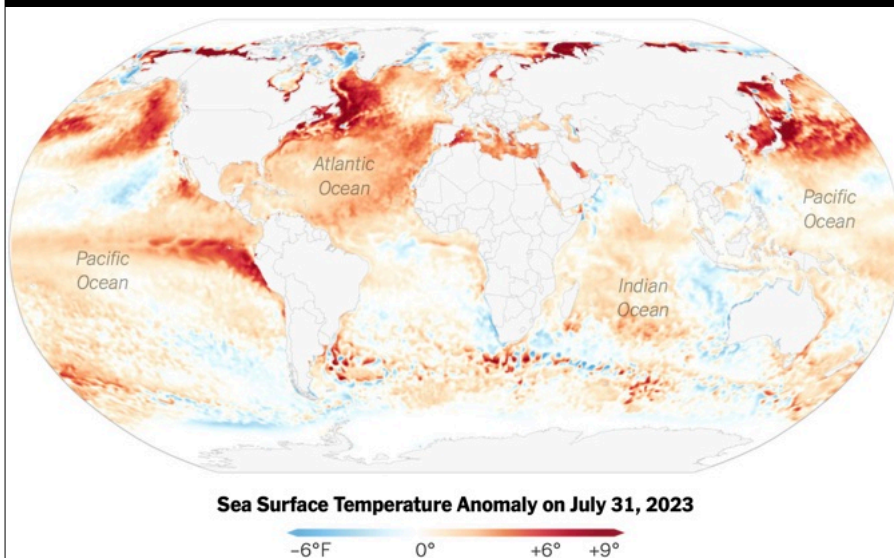


3.5 inches in past 25 years, and accelerating

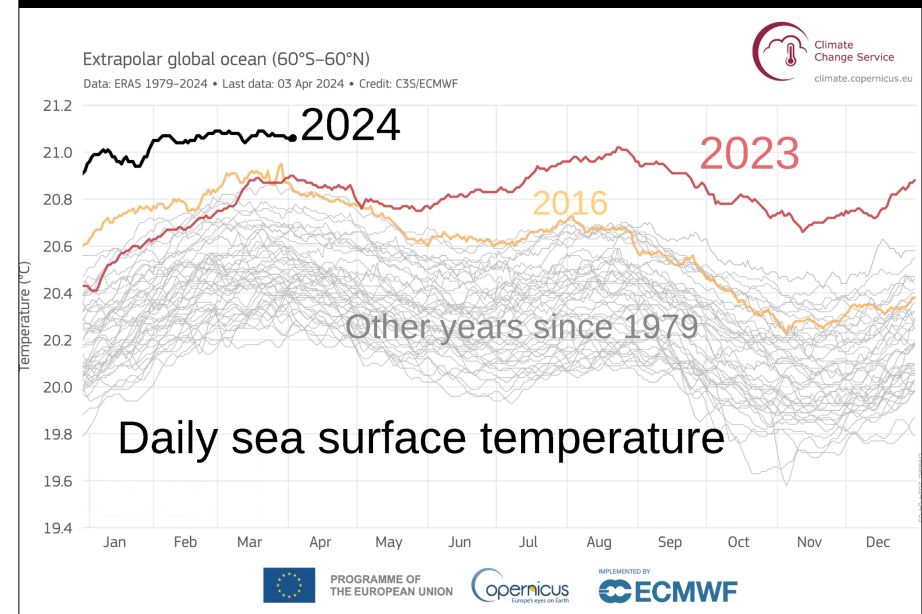
The ocean is absorbing most of the heat

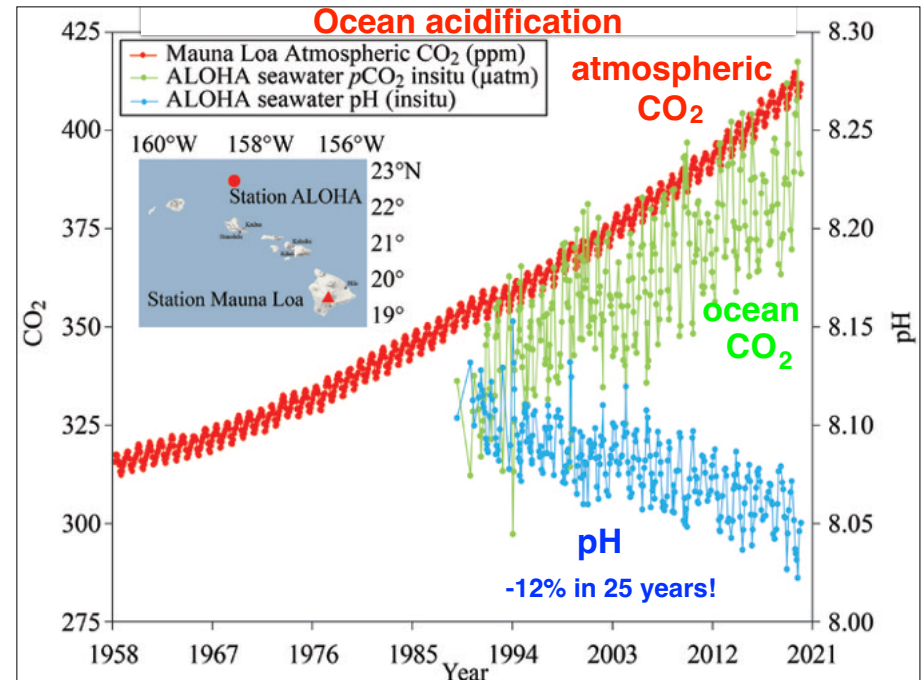
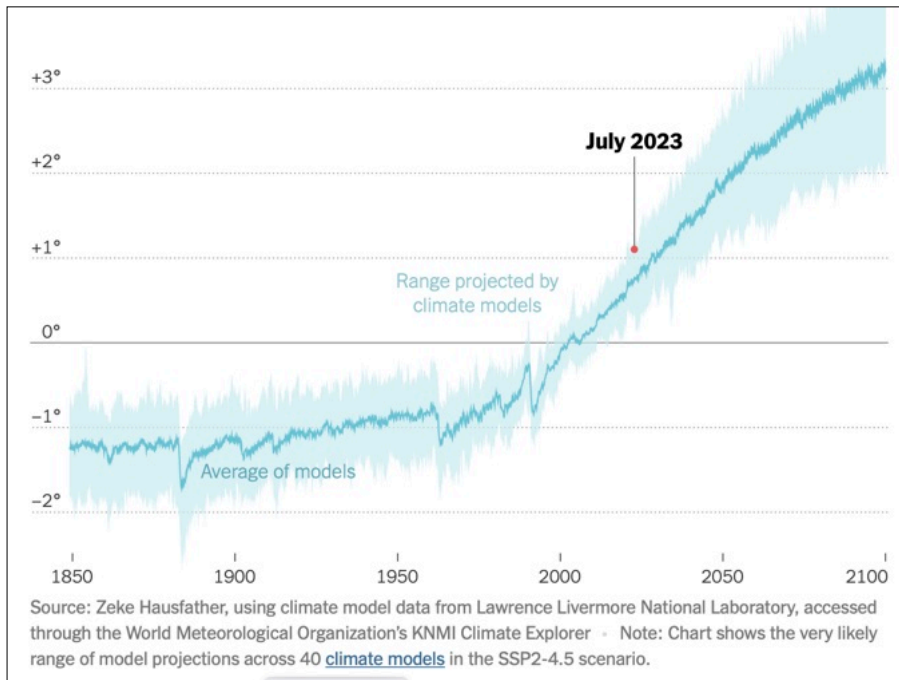


Heat is accumulating in the ocean unevenly



Heat accumulation in the ocean is accelerating





Chapter 7: Modeling the climate

- the models are highly complex
- they can be tuned to explain the past quite well
- they all predict a strongly warming world
- there are recent changes that are mildly surprising

What we have accomplished

- Chapter 1: Astronomical factors
- Chapter 2: Earth's surface and atmosphere
- Chapter 3: The source of CO₂
- Chapter 4: The temperature record
- Chapter 5: Paleoclimate
- Chapter 6: Energy Balance and Feedback
- Chapter 7: Modeling the climate

Chapter 8: Fictions and conclusions

Fiction

“97% of climate scientists agree that human activity is warming the planet”

24,210 papers published in refereed journals in 2013 & 2014
(*global climate change* or *global warming* in title or abstract)

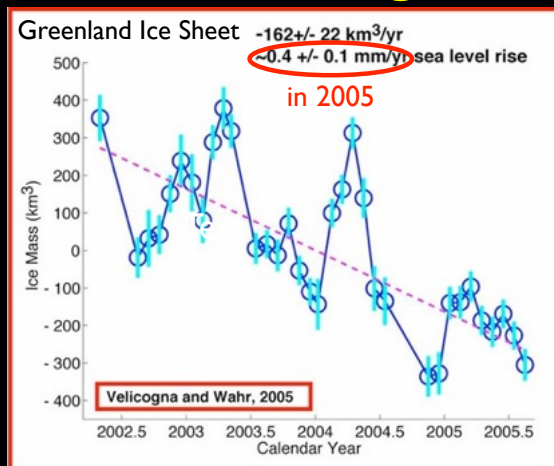
5 papers reject anthropogenic warming
(two of these by the same author; only one has a citation)

Thus, only 4 out of 69,406 authors deny climate change

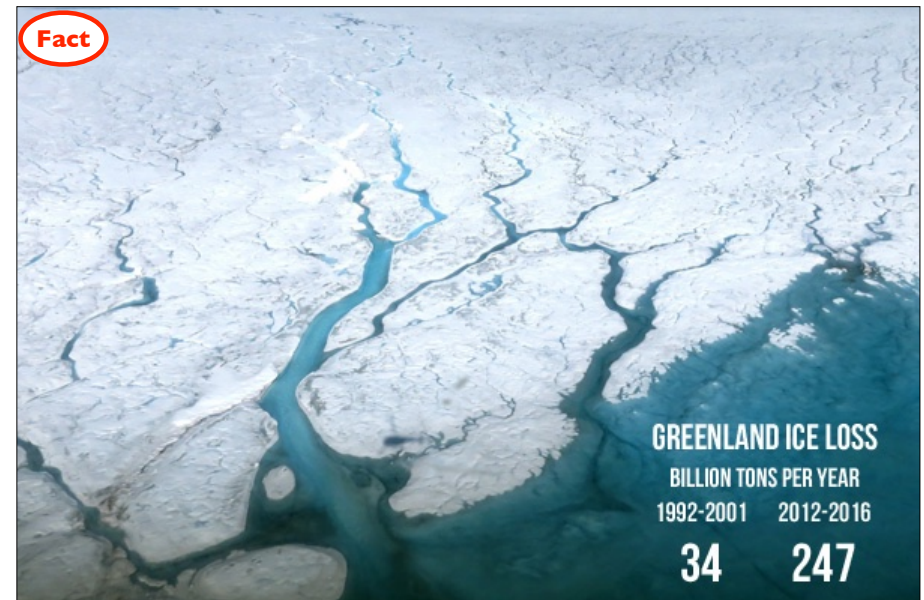
99.994% of climate scientists agree that humans are warming the planet

(Fiction)

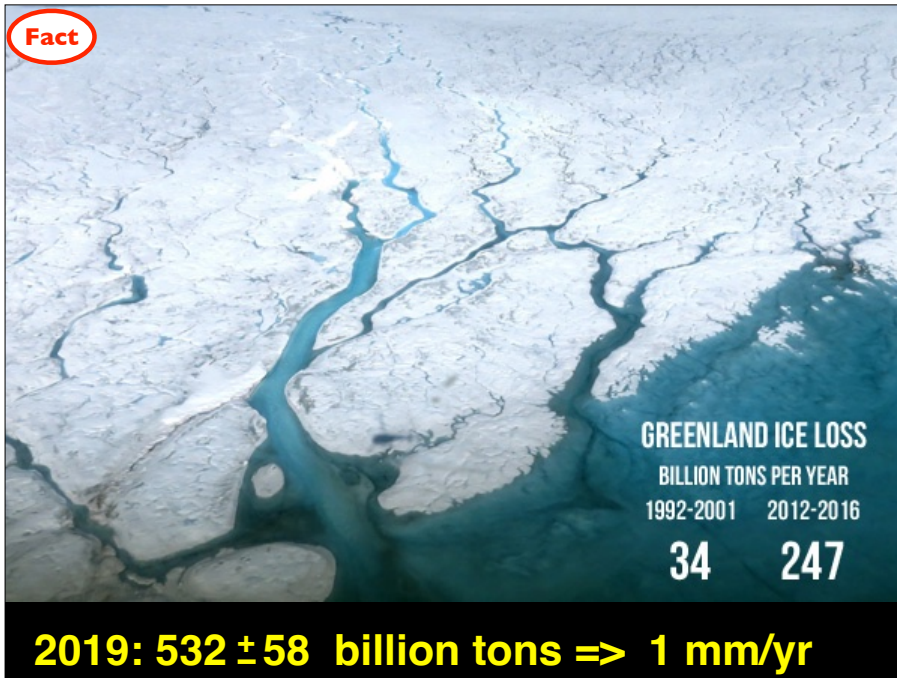
Melting glaciers are raising sea levels



Fact



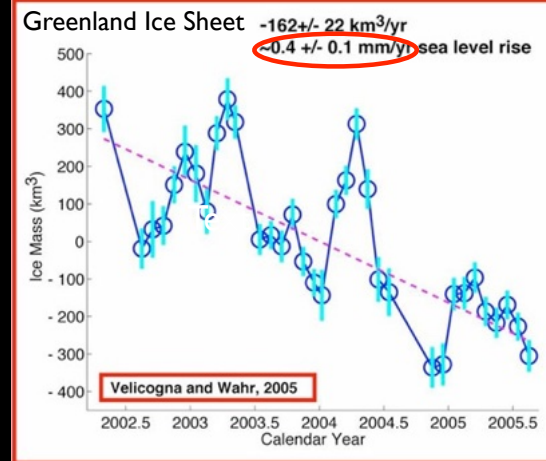
Fact



(Fiction)

Melting glaciers are raising sea levels

Total sea level rise is > 4 mm/yr



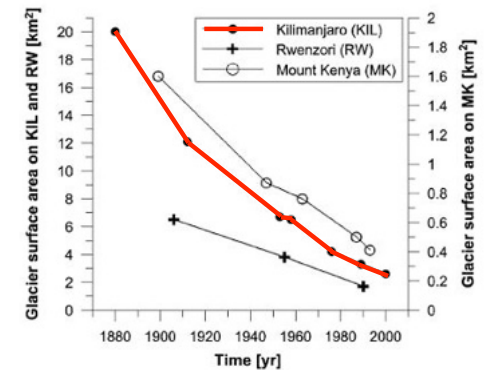
$\sim 50\%$ of sea level rise is due to thermal expansion of oceans

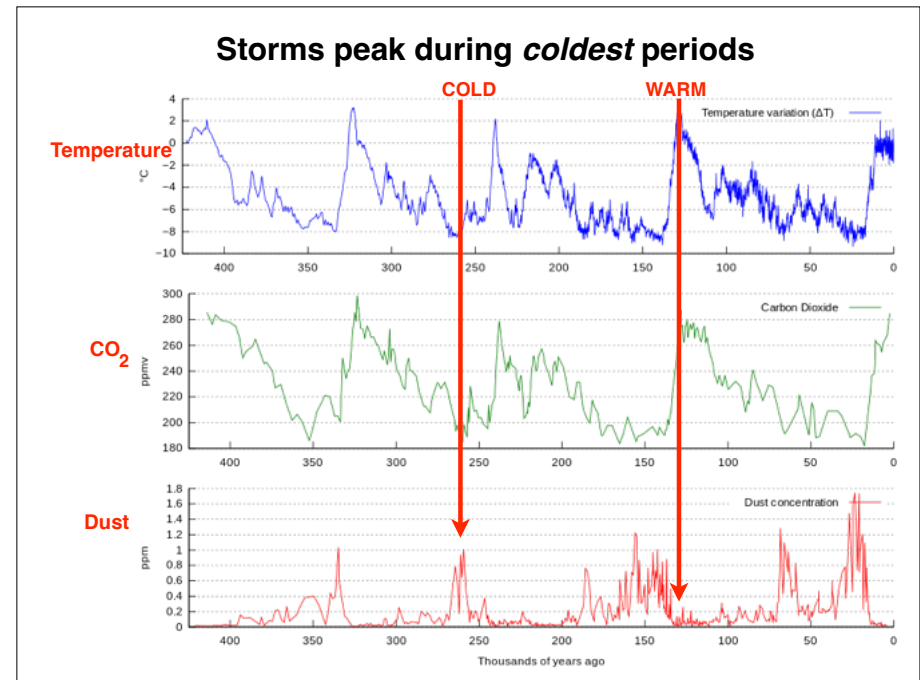
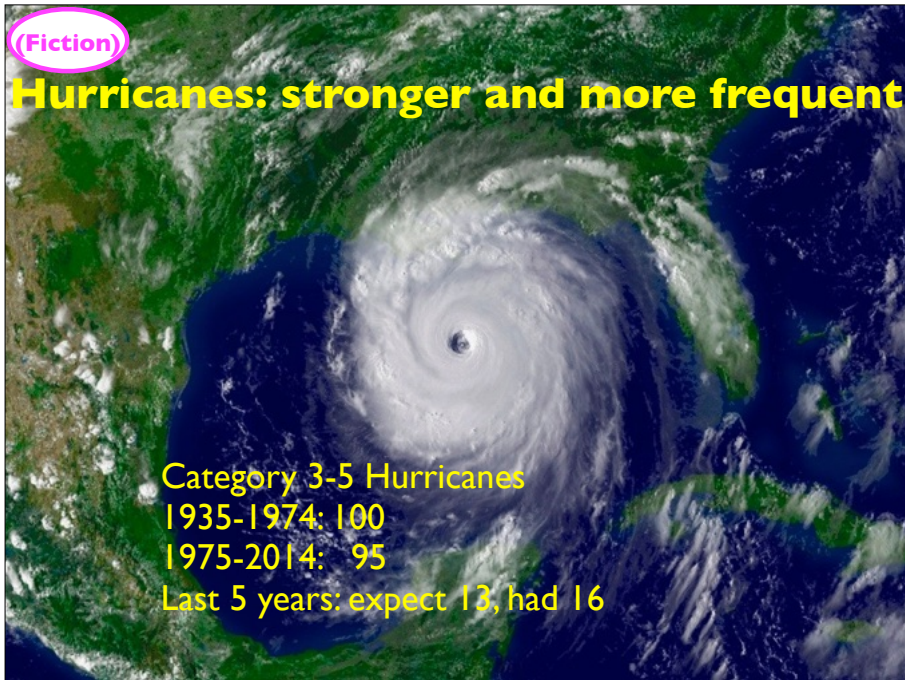
(Fiction)

The Snows of Kilimanjaro

1993

2000





Chapter 9: Where we are and where we're going

8.2 billion people

300,000 years to get to 1 billion people
 11 years to go from 7 billion to 8 billion

Human biomass is ~100 x that of any large animal that has ever existed on Earth (E.O. Wilson)

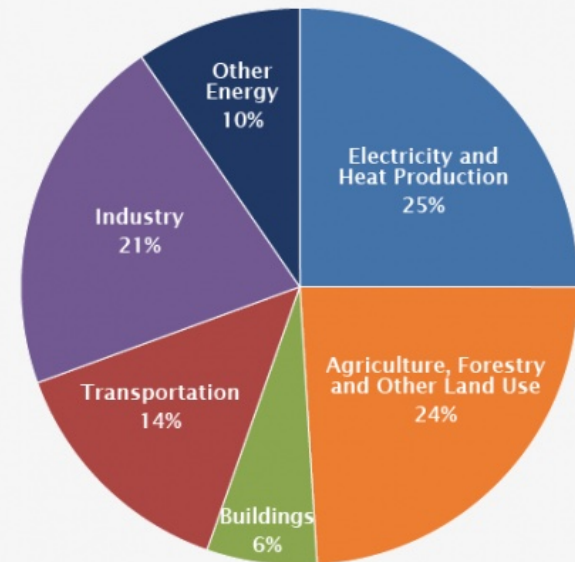
Basic Energy Facts

Our metabolism: 100W
 Our (US) energy consumption: 10,200W
 Average per capita energy use: 2,575W

World power consumption: $20 \times 10^{12} \text{ W}$
20 TW

(Sunlight reaching Earth: $1.75 \times 10^{17} \text{ W}$)
 175,000 TW

Global Greenhouse Gas Emissions by Economic Sector

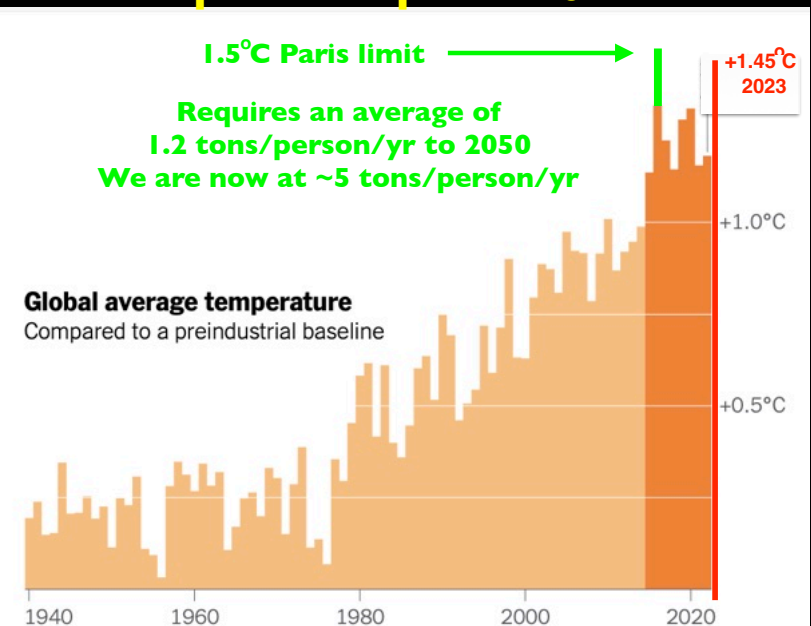


**It isn't
just one
thing,
it is
everything**

December 12, 2015



T compared to pre-industrial T



Paris Pledges tons of CO₂ per person by 2030

US: 15.96

China: 9.06

India: 4.53

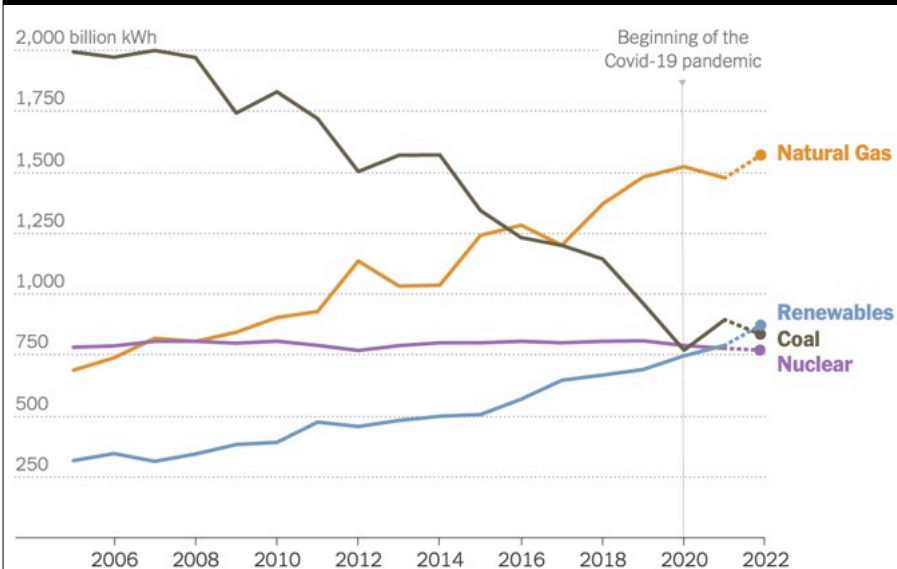
EU: 6.43



Current US total energy use

	2016	2022
Fossil Fuel:	79%	79%
Nuclear:	8.4%	8.0%
Biomass:	4.8%	4.8%
Hydro:	2.5%	2.3%
Wind:	2.1%	3.8%
Solar:	0.6%	1.8%
Other:	1.2%	0.3%

US trends: coal down, but gas up



Paris Pledges (tons CO₂ per capita)

US: ~~15.96~~

China: 9.06

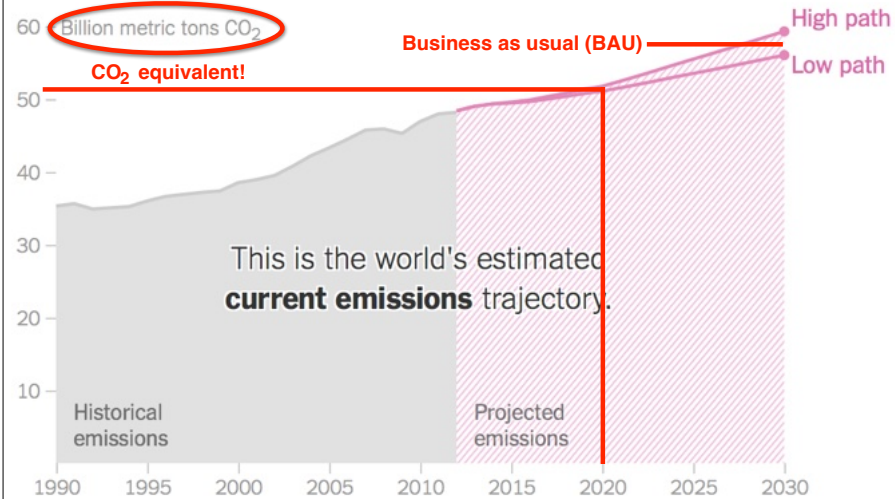
India: 4.53

EU: 6.43

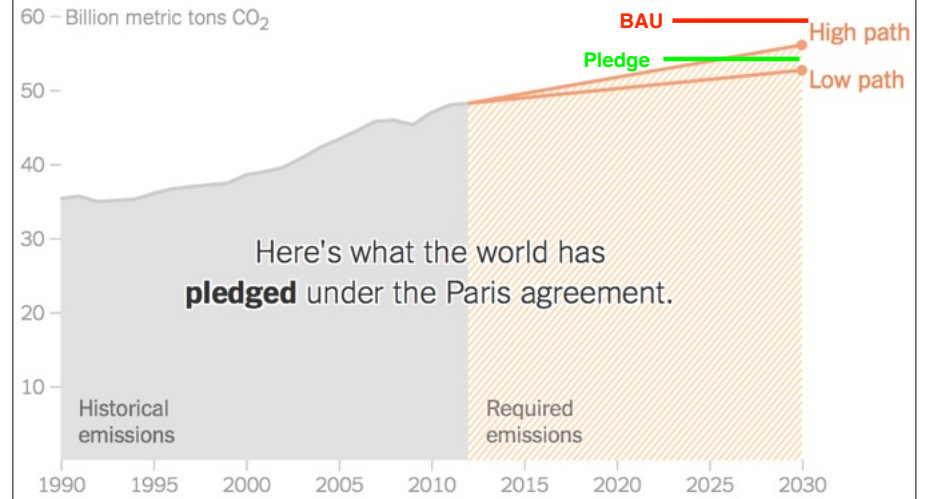


193 countries in, 3 out

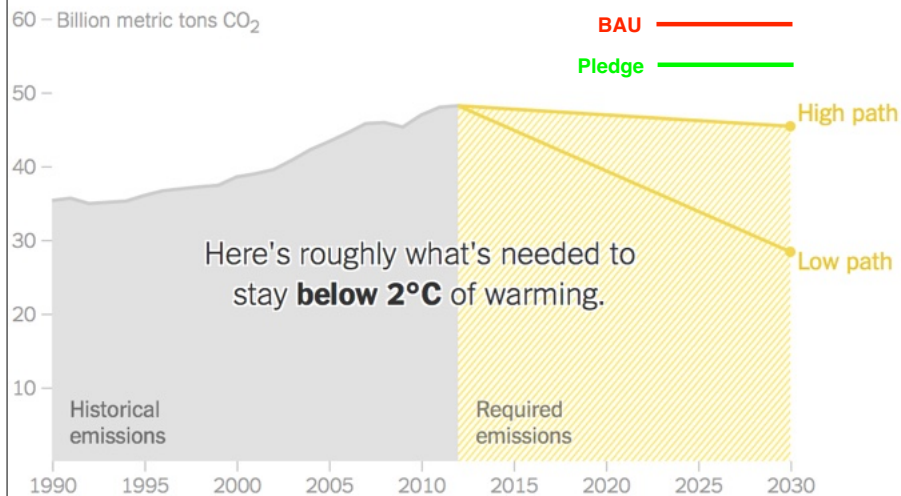
Global Total — Business as usual



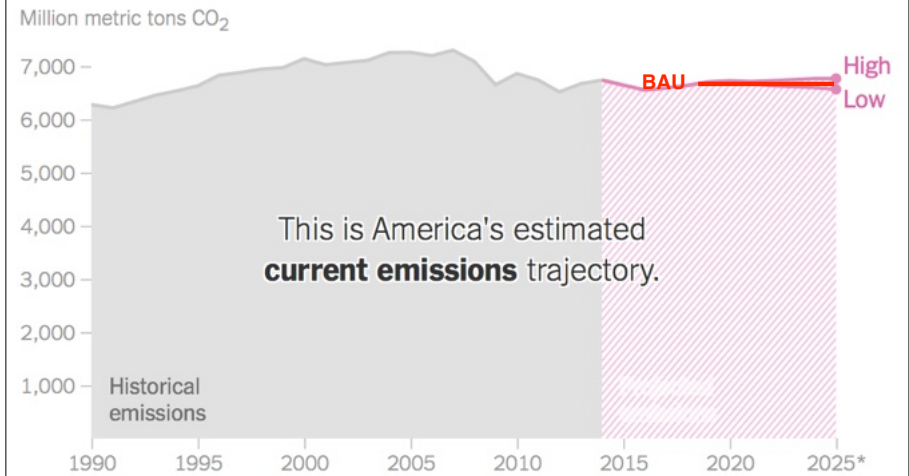
Global Total Pledged

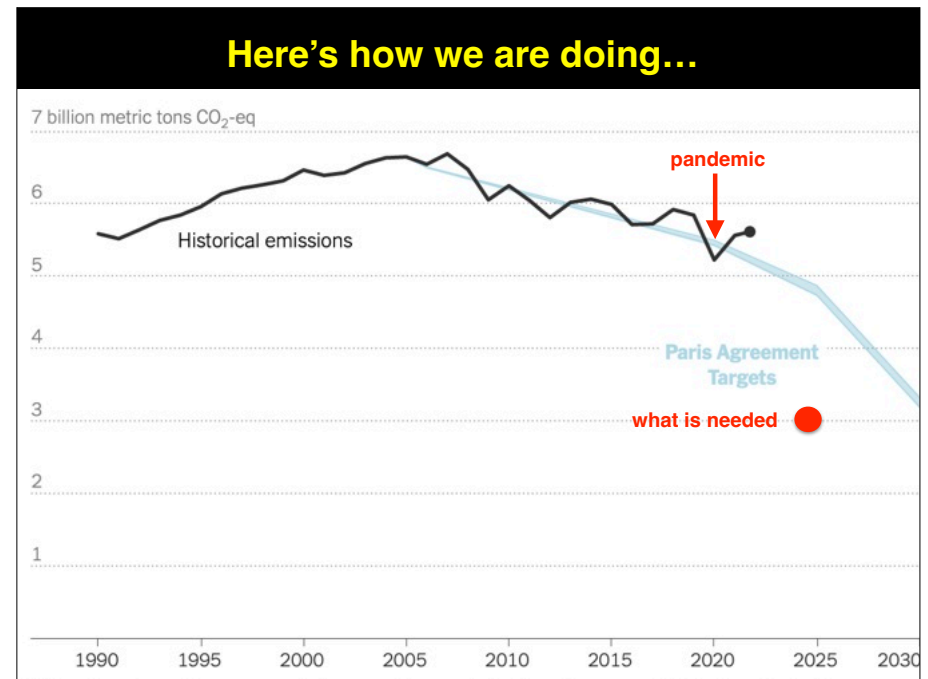
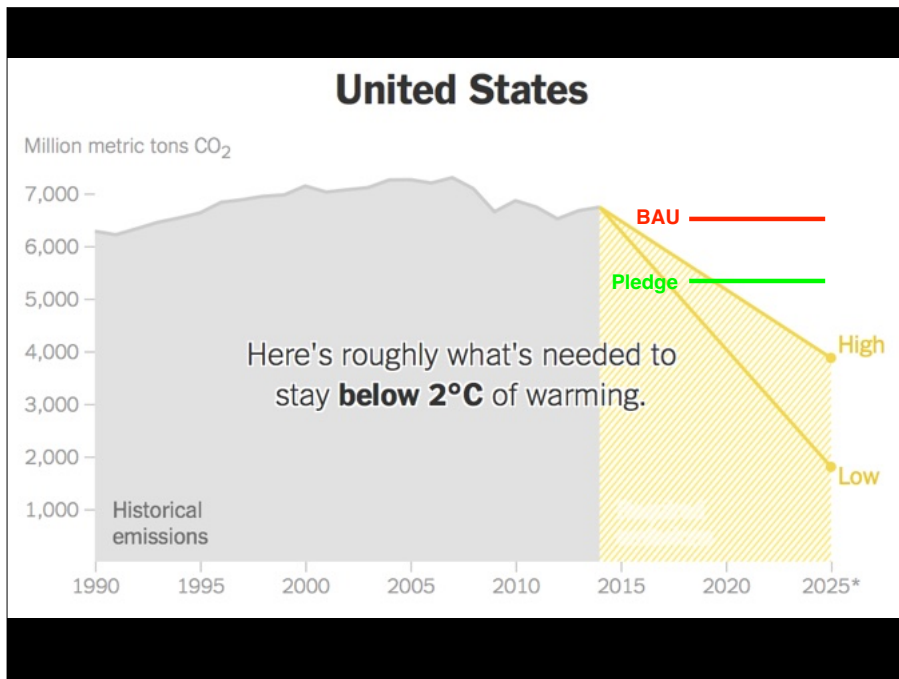
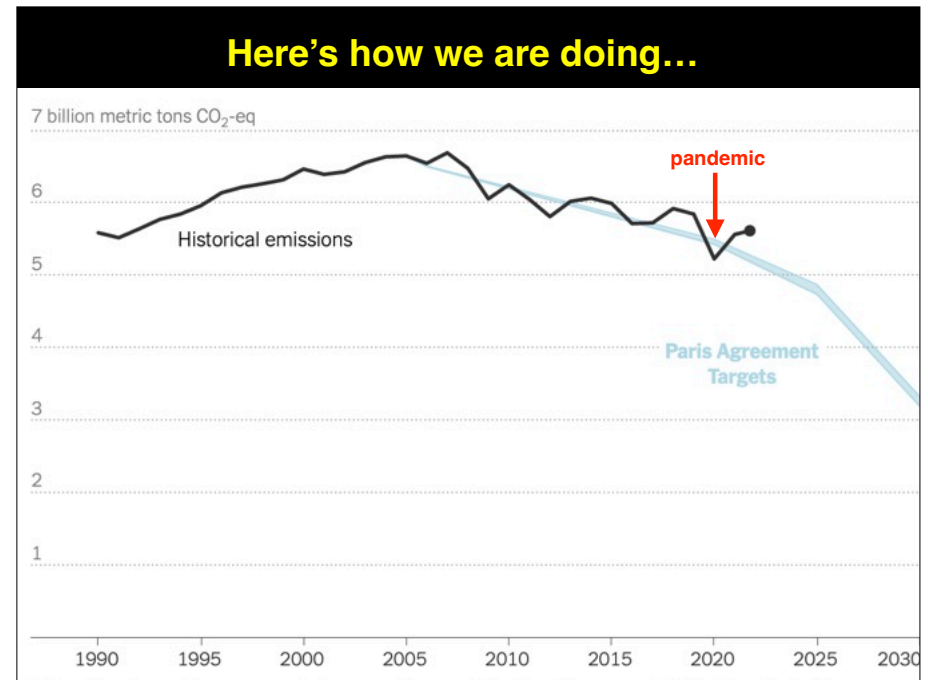
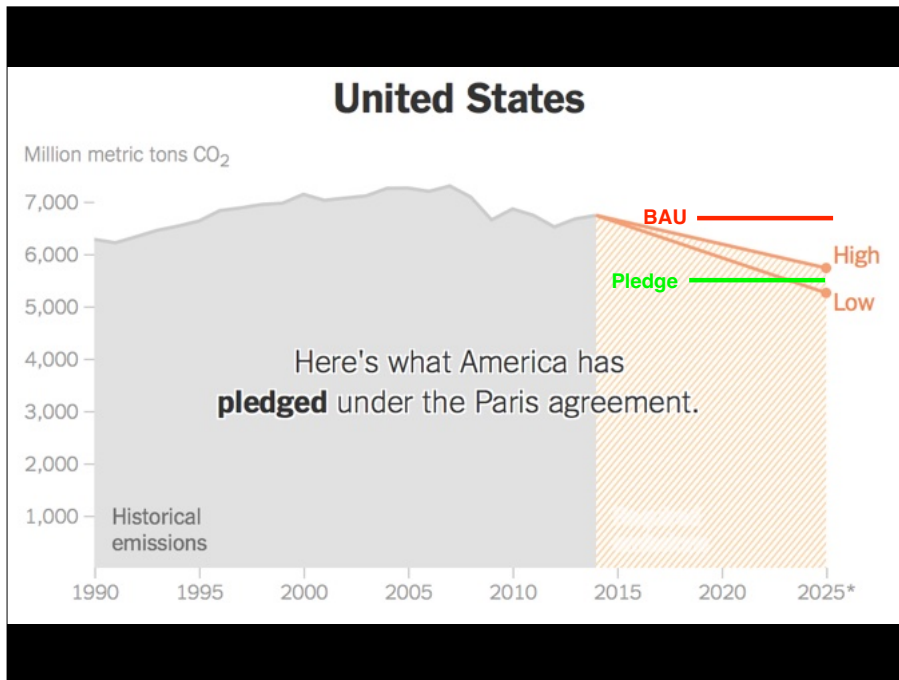


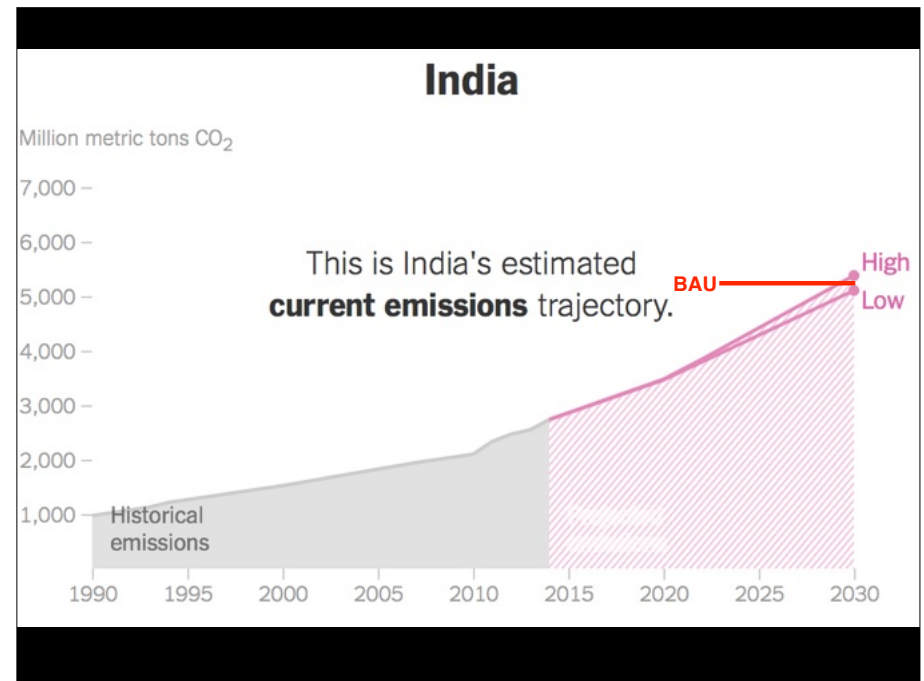
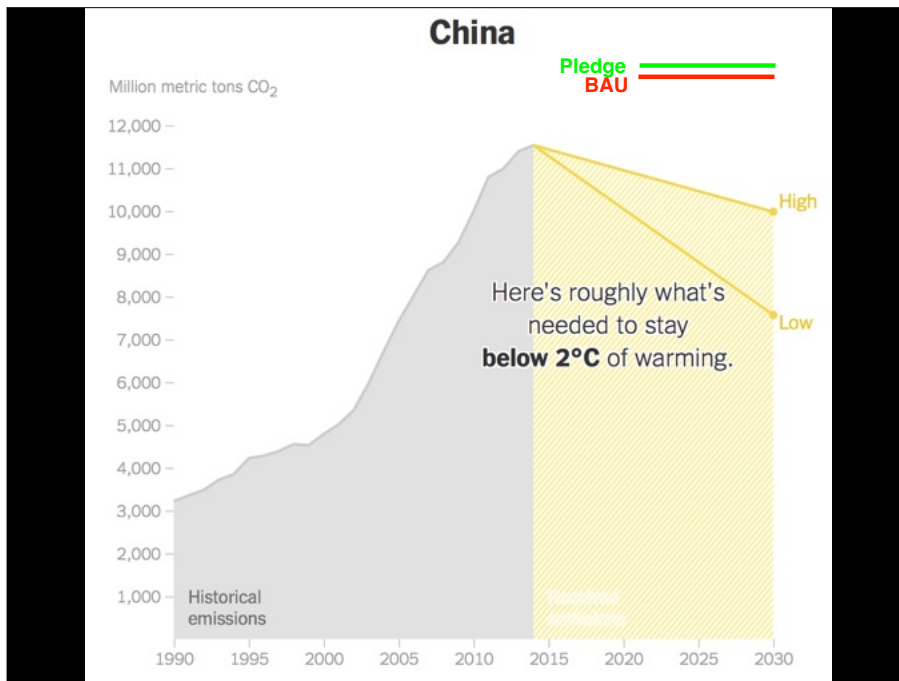
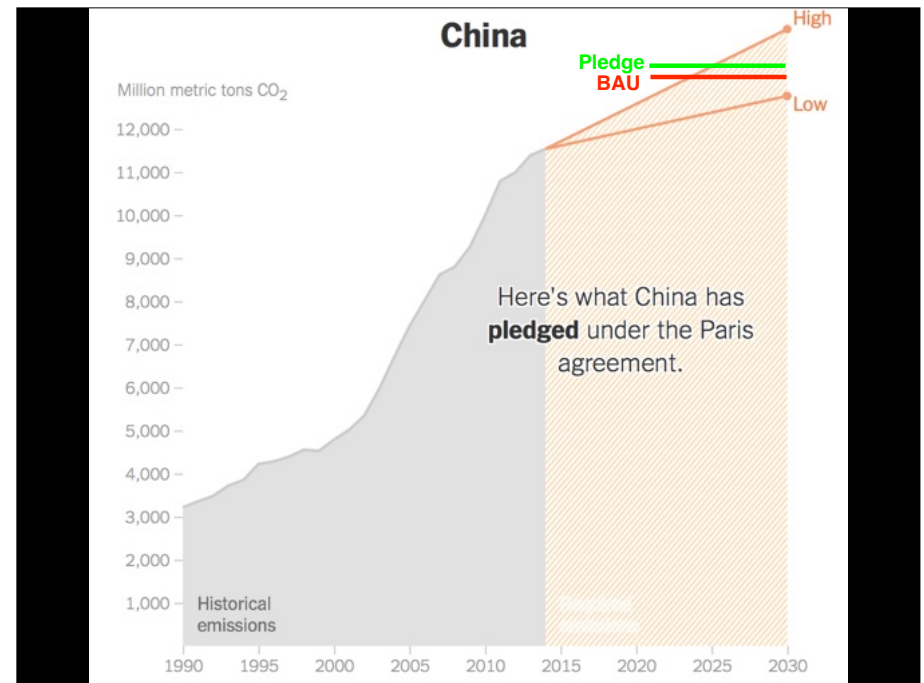
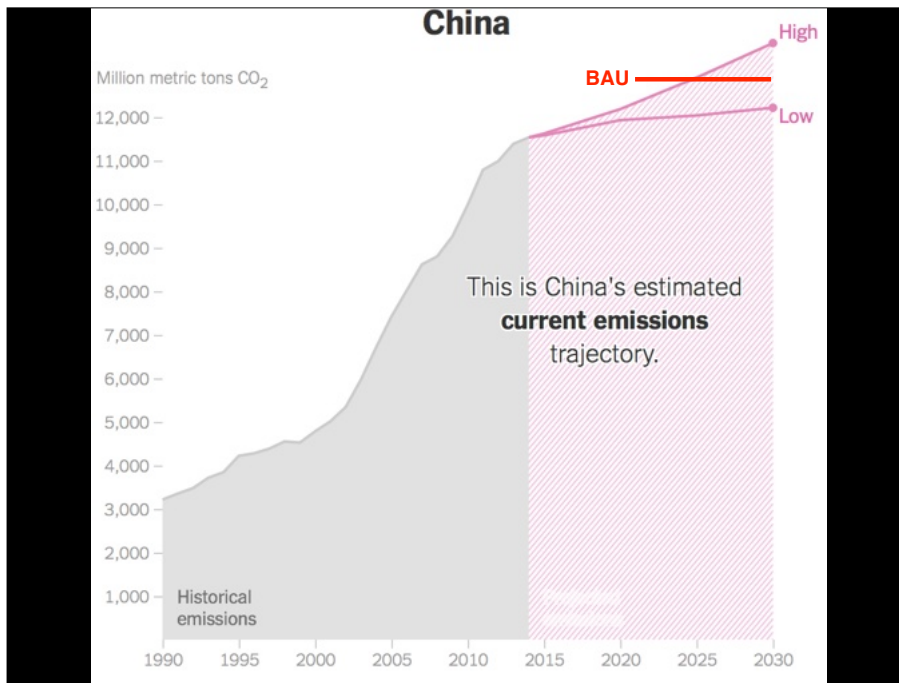
Global Total Required



United States

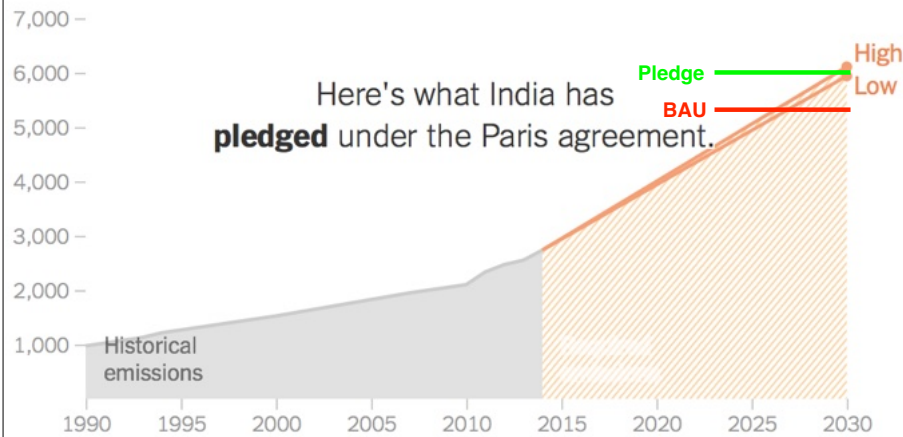






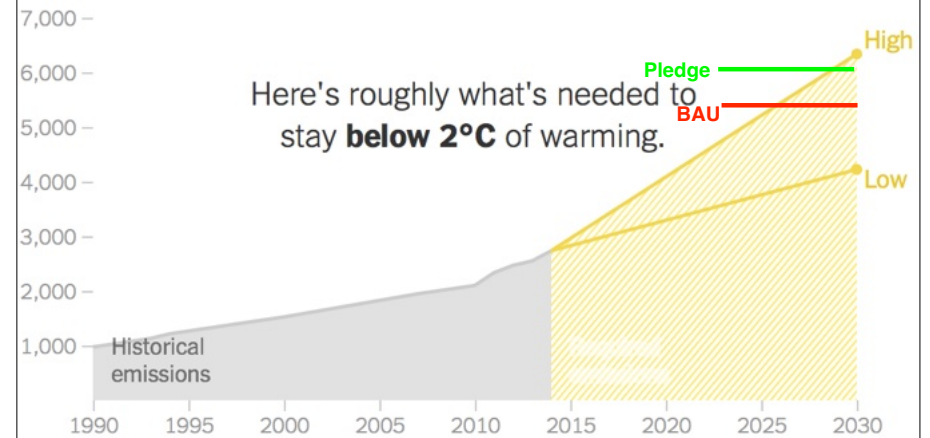
India

Million metric tons CO₂



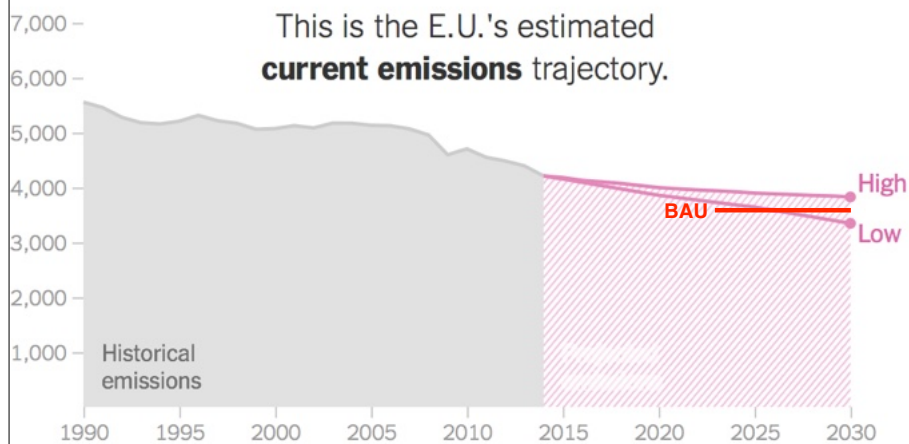
India

Million metric tons CO₂



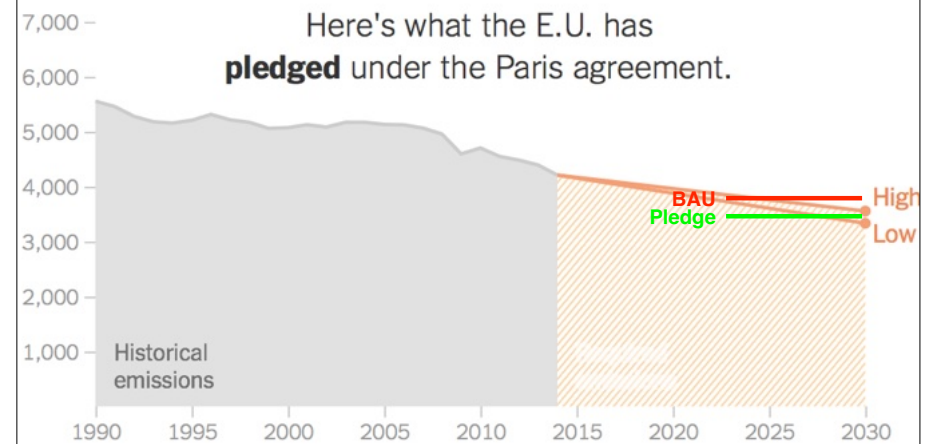
European Union

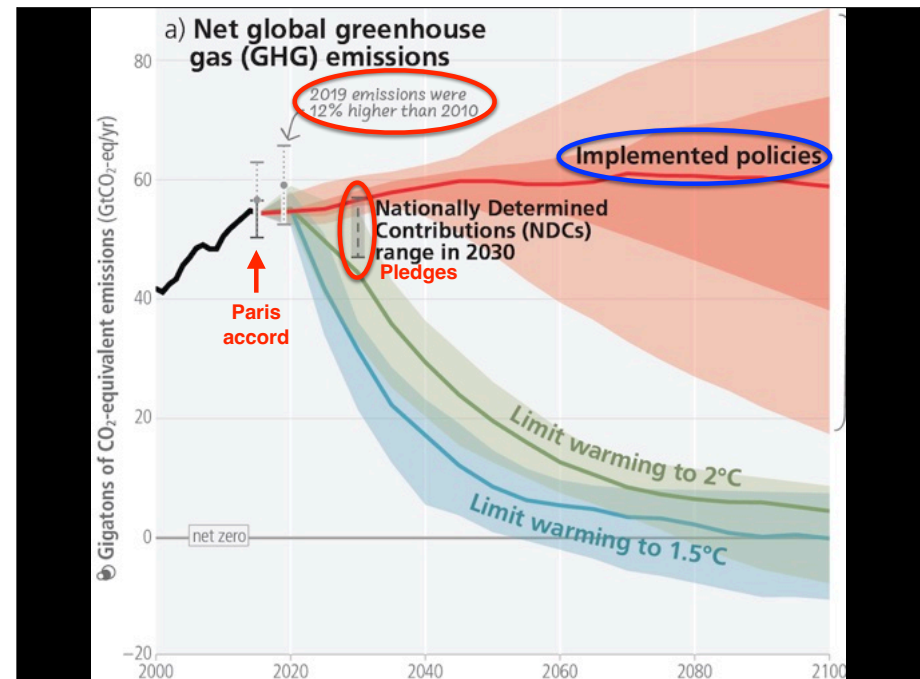
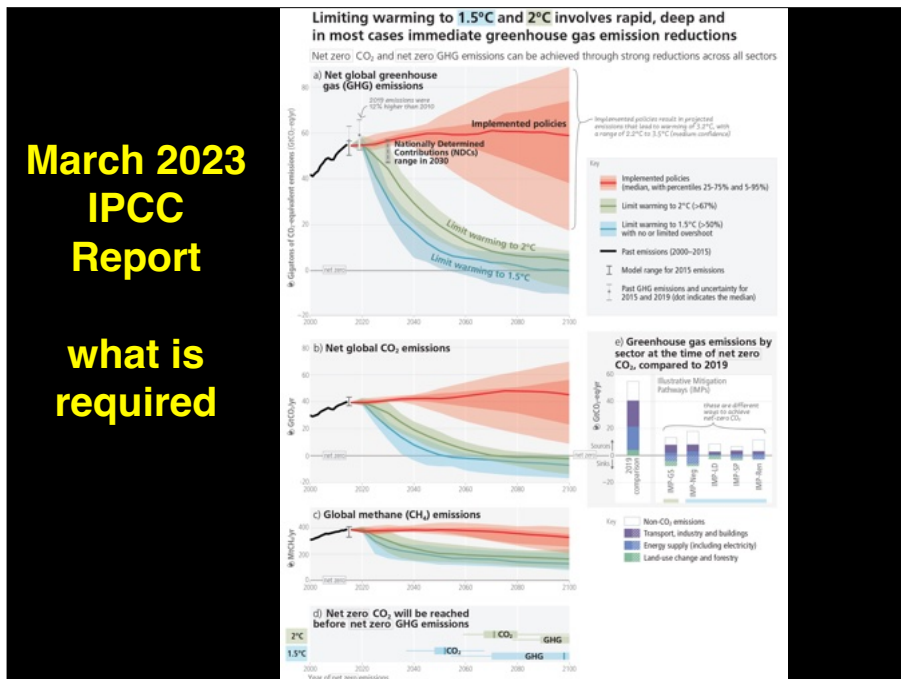
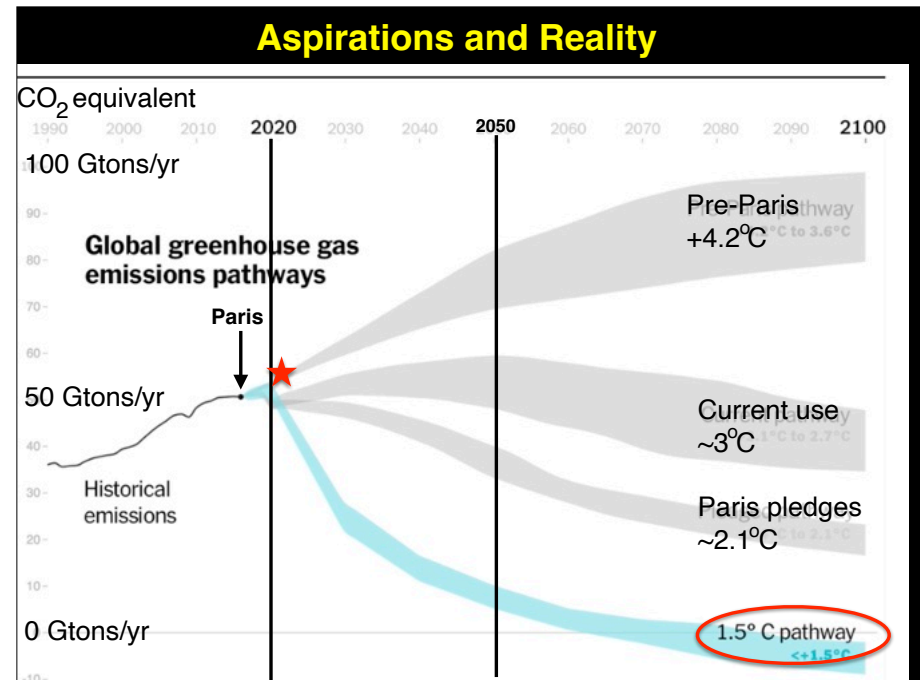
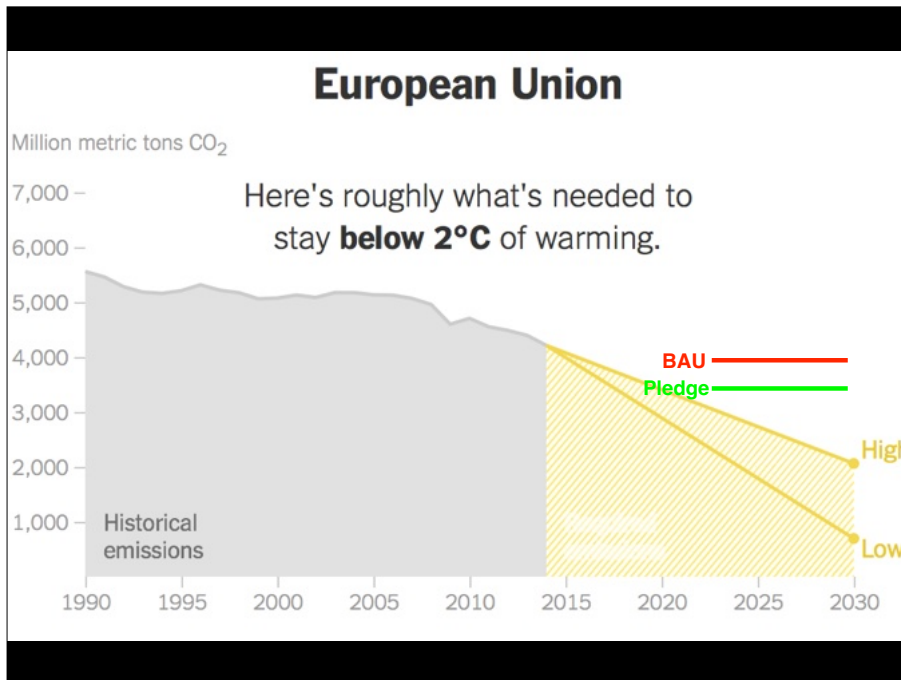
Million metric tons CO₂



European Union

Million metric tons CO₂





As of 2020

2/193 countries

Morocco and The Gambia
(0.5% of the world's population)

have met their 2015 pledges

Of the top ten emitters only

India
(Solar+hydro = 34% vs US = 3.5%)
is close to on-track for +2° C

It is not about the future for many people;
it's about the present



for the Choctaw people from
Isle de Jean Charles, Louisiana

It is not about the future for many people;
it's about the present



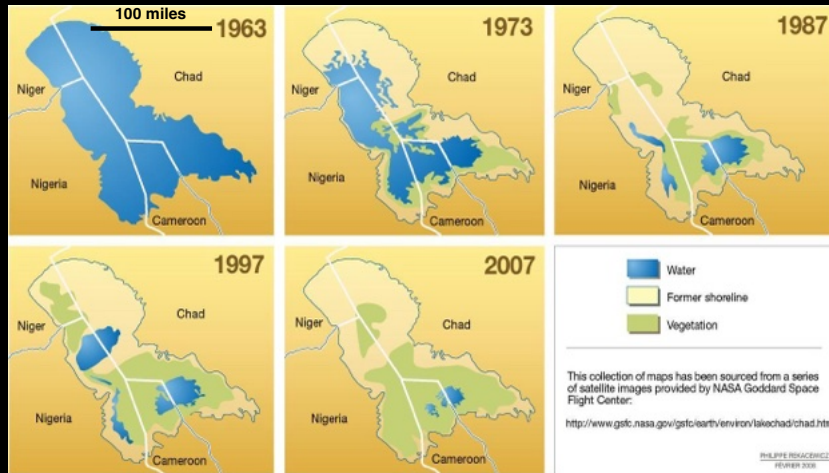
for 600 Inuiptat people from
Shishmaref, Alaska

It is not about the future for many people;
it's about the present



for the 1 million drought refugees from
northeastern Syria

**It is not about the future for many people;
it's about the present**



**for 2.5 million displaced people from
Niger, Chad, Nigeria, and Cameroon**

**It is not about the future for many people;
it's about the present**



**for 18 million people (within decades)
in Bangladesh**

**It is not about the present for some people;
it's about the future — so why worry?**



**if you are among the people
who fund our politicians**

Last Thanksgiving:

“Today, I am thankful I live a high-carbon lifestyle and wish the rest of the world could too. Energy poverty = poverty
#decarbonization is dangerous and deadly.”

— Jason Isaac,
Director, Life : Powered initiative
Texas Public Policy Foundation

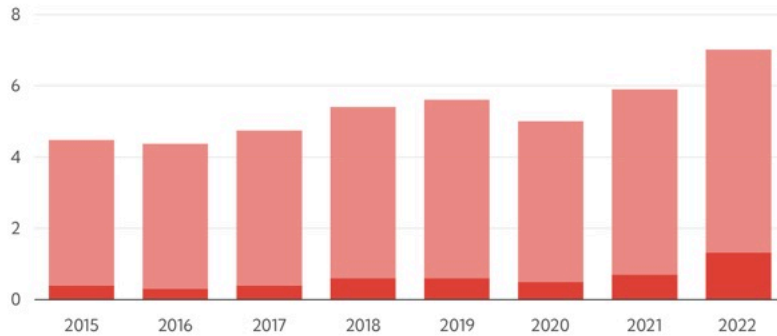
“Believing in Climate Change is a form of Paganism”

— Kathleen White,
Fellow, Texas Public Policy Foundation and
Trump nominee to lead
Council on Environmental Quality

\$7,000,000,000,000 for fossil fuels and rising

Fossil fuel subsidies topped \$7 trillion last year
(total fossil fuel subsidies, trillions of USD)

■ Explicit subsidies ■ Implicit subsidies



Source: IMF staff calculations.

Note: Figures from 2019 onwards use projections for fuel use. Explicit subsidies: undercharging for supply costs. Implicit subsidies: undercharging for environmental costs and forgone consumption taxes, after accounting for preexisting fuel taxes and carbon pricing.

IMF

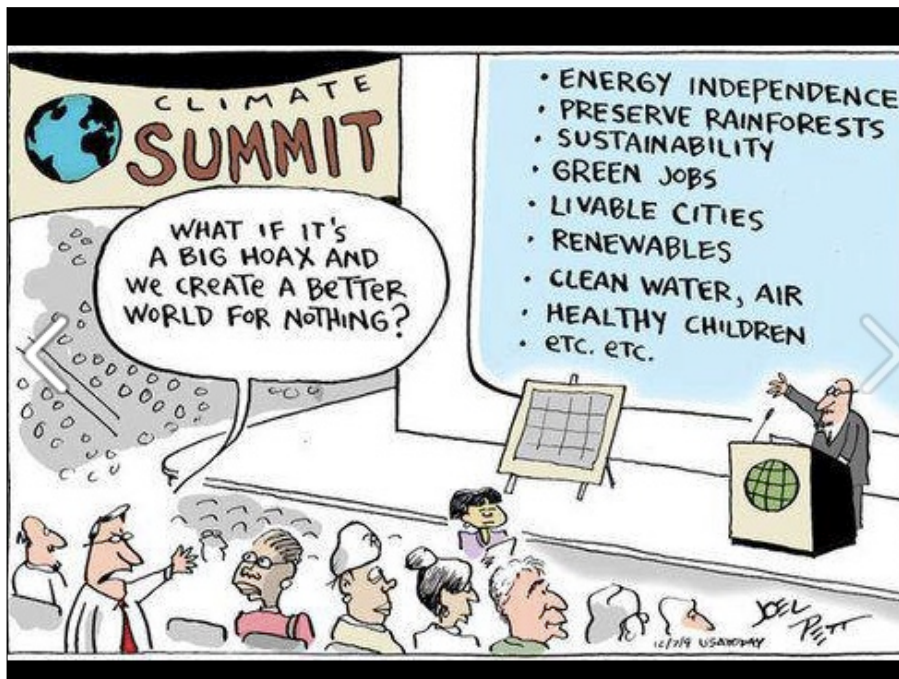
\$619,000,000,000 (9%) investment in renewable energy

Frank Luntz, testifying before a Congressional Committee, June 2019:

“I’m here before you to say that I was wrong in 2001.

Rising sea levels, melting ice caps, tornadoes, and hurricanes more ferocious than ever. It is happening.”

He advocated “a no-regrets strategy,” suggesting legislation would lead to cleaner air, cleaner water, less dependence on foreign fuels, enhanced national security, and more innovation in our economy. “And that’s if the scientists are wrong. If the scientists are right, we get all of those things and begin to solve what could be the most catastrophic environmental problem that any of us have ever faced ... That’s why it’s the right thing to do.”



Things you are told to worry about:

- stronger and more frequent hurricanes
- rising sea levels
- hotter summers and deaths from heat stroke
- the demise of the polar bears

Things that I worry about much more:

- spreading tropical diseases
- CO₂ from boreal bogs
- biodiversity collapse
- fresh water exhaustion

(and too many home runs: 500 extra from global warming 2010-2019)

Ground water exhaustion is widespread



This is *not* an insoluble problem

- solar energy reaching Earth is >8000 times our current total global energy use
- plentiful uranium and thorium for nuclear power is available as a bridge
- individual actions help but are not a global solution



This is not the first time living creatures
have fundamentally altered the Earth

This is not the first time living creatures
have fundamentally altered the Earth

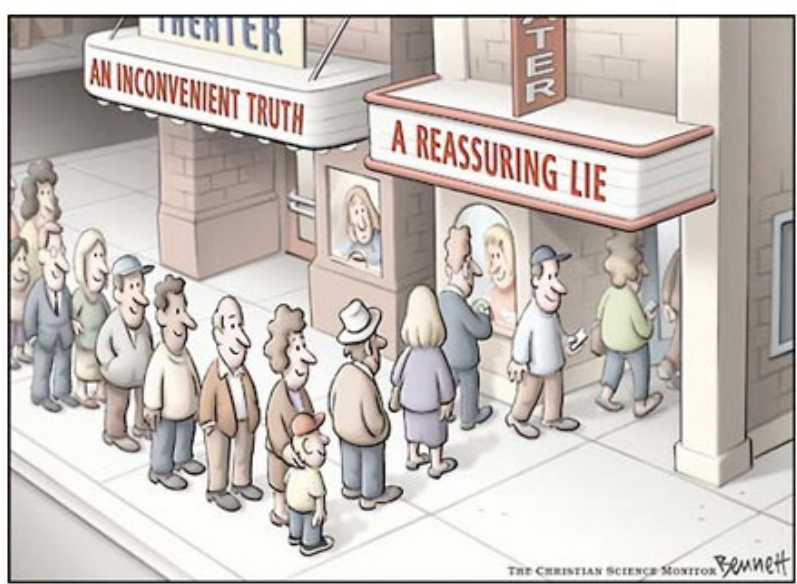
It is just the first time they could make a
decision not to

This is not the first time the Earth's
climate has changed

This is not the first time the Earth's
climate has changed

It is just the first time the change may be
partially within our control

YOU MUST CHOOSE WHICH WAY TO GO



Stephen Markley in New York Times:

"Climate is not just another issue. I do not deny that we live in a complex and precarious world or that our consciences are torn by a web of domestic challenges and geopolitical upheavals. But we are in denial if we do not recognize that climate is the crisis that will define this century and if we fail, the entire human future. Our fossil fuel system is driving the planet to a set of conditions humans have never experienced, where even the imagination of novelists will fail us.

...the climate crisis is the foundation on which we can build a more just, equitable and prosperous world. Every election is precious, every ballot we cast a moral record of what we did in this crucial historical moment. Do not sit on your hands, do not deny the stakes, do not waste that vote."

This is not the first time Earth's future is uncertain

This is not the first time Earth's future is uncertain

It is just the first time that "future" is a meaningful concept

There is no plan(et) B



See you next week for the Frontiers Finale

